Efficient Estimation of Nonlinear DSGE Models

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Abstract

This paper introduces a computationally efficient and accurate method for estimating nonlinear DSGE models by integrating model solution and filtering. We combine local linear solution techniques with a time-varying Kalman filter, yielding the optimal linear filter for this solution method. We benchmark the method using the canonical Diamond-Mortensen-Pissarides (DMP) model and find that it matches or exceeds the accuracy of global solution methods combined with nonlinear filters. Moreover, it reduces computational time by several orders of magnitude relative to existing estimation approaches. A key feature of the framework is that it produces state-dependent impulse responses, computed at each point in time from filtered states. Applying the method to a New Keynesian model with labor market frictions, we find that endogenous variation in job creation profitability induces strong state dependence in monetary policy transmission. In both the estimated model and reduced-form state-dependent local projections, the Phillips Multiplier is approximately 50 percent larger during expansions than during recessions.

Keywords: numerical methods, Taylor projection, labor search, full-information estimation

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1 Introduction

Nonlinearities play a fundamental role in macroeconomic dynamics, and recent empirical evidence increasingly underscores their importance. Several studies suggest that the Phillips curve is nonlinear or exhibits a time-varying slope, while a growing literature documents that the transmission of economic shocks is state-dependent and varies with both the magnitude and the sign of the shock.[1] While macroeconomic models often incorporate such nonlinearities, whether these features meaningfully affect model predictions depends crucially on the choice of solution and estimation method. The prevailing approach to full-information estimation of nonlinear Dynamic Stochastic General Equilibrium (DSGE) models — global solution methods combined with nonlinear filters — remains computationally prohibitive.[2] As a result, researchers frequently resort to computationally simpler methods that retain linearity, despite widespread recognition that linear approximations can yield substantially different parameter estimates.[3]

Methodological Contribution We develop a computationally efficient full-information estimation method for nonlinear DSGE models. Whereas traditional estimation approaches treat the model solution and likelihood evaluation as separate steps, we propose an integrated framework that embeds the model solution directly within the filtering step.

The key insight builds on recent advances in local linear solution methods for DSGE models (Fernández-Villaverde and Levintal, 2018; Levintal, 2018; Mennuni, Rubio-Ramírez, and Stepanchuk, 2024). We show that models solved using local linear approximations can be mapped into a time-varying linear state-space representation. This structure enables a filtering procedure in which, at each predicted state, a new set of local linear policy functions is computed. We then filter the model using a time-varying Kalman filter, generating updated forecasts at each step where the model is re-solved. This approach is both computationally efficient, owing to the speed of local linear methods, and optimal within the solution class.

Our method applies to a broad class of DSGE models and allows researchers to capture key nonlinearities in a computationally tractable way. Moreover, local linear methods preserve state dependence, meaning that shock propagation varies with the state of the economy, and, despite the local linearity, the solution is not certainty equivalent.

[1] For example, Benigno and Eggertsson (2024); Gitti (2024); Hazell, Herreno, Nakamura, and Steinsson (2022); Smith, Timmermann, and Wright (2023); Stock and Watson (2020) examine time variation in the Phillips curve; Tenreyro and Thwaites (2016) studies state dependence in monetary policy transmission; and Barnichon, Debortoli, and Matthes (2022) analyzes sign dependence in government spending multipliers.

[2] For example, Gust, Herbst, López-Salido, and Smith (2017) report that a single likelihood evaluation requires six minutes on a supercomputer using parallelization.

[3] See, for example, Fernández-Villaverde and Rubio-Ramírez (2005), An and Schorfheide (2007), Amisano and Tristani (2010), and Gust et al. (2017), as well as Judd, Maliar, and Maliar (2017) on large approximation errors associated with perturbation methods.

We benchmark the proposed estimation method on two canonical models to highlight its accuracy and computational efficiency. First, we consider the Diamond-Mortensen-Pissarides (DMP) model,[4] a framework characterized by pronounced nonlinearities (Petrosky-Nadeau and Zhang, 2017). Using a global solution of the DMP model as the data-generating process, we demonstrate that our approach matches or exceeds the accuracy of the true global solution combined with either the Particle Filter or the Extended Kalman Filter, but at a fraction of the computational cost. Second, we estimate the canonical three-equation New Keynesian (NK) model (see, e.g., Gali, 2015; Herbst and Schorfheide, 2016). Our method obtains 100,000 posterior draws in less than eight minutes—less than twice the time required for estimating a linearized version using Dynare,[5] and in stark contrast to nonlinear estimation via second-order perturbations and particle filtering, which is approximately 200 times slower (An and Schorfheide, 2007). Solving the model globally would be even more computationally demanding.

Main Application As a main application, we apply the proposed estimation framework to assess state dependence in monetary policy transmission within a New Keynesian model augmented with Diamond-Mortensen-Pissarides (DMP) labor market frictions. In labor search models, the profitability of job creation—the flow surplus—is a key determinant of how technology shocks propagate through the economy (Ljungqvist and Sargent, 2017). We show that this same mechanism extends to monetary policy shocks. A central insight is that while flow surplus is fixed at its steady-state value in linearized models, it evolves endogenously in nonlinear solutions, generating time-varying shock transmission.

A key advantage of our framework is that it permits filtering of latent economic states, allowing monetary policy transmission to be evaluated conditional on the current state of the economy. In particular, it facilitates the computation of state-dependent impulse response functions (IRFs). Our model estimates reveal substantial variation in IRFs, with the effects of monetary shocks on output, unemployment, and market tightness varying by as much as a factor of two depending on economic conditions.

A particularly policy-relevant implication is the resulting variation in the Phillips Multiplier, which quantifies the inflation-unemployment trade-off. On average, we estimate the Phillips Multiplier at 0.15, consistent with Barnichon and Mesters (2021). Crucially, our framework allows this object to be tracked dynamically. We find a cyclical pattern: during recessions, when marginal hiring costs fall, the Phillips Multiplier declines to values as low as 0.11. During expansions, when hiring costs rise, it increases, approaching 0.18.

To validate these model-based findings, we present reduced-form empirical evidence based on state-dependent local projections (Cloyne, Jordà, and Taylor, 2023). Using two proxies for

[4] Diamond (1982), Mortensen (1982), and Pissarides (1985).

[5] Adjemian et al. (2024).

flow surplus—unemployment and corporate profits—we document cyclical variation in both the Phillips Multiplier and the transmission of monetary shocks, consistent with the model's predictions.

More broadly, our estimated NK-DMP model demonstrates that several nonlinear features emphasized in recent empirical work—such as state-dependent monetary policy transmission (Tenreyro and Thwaites, 2016) and cyclicality in the Phillips curve slope (Benigno and Eggertsson, 2024; Gitti, 2024)—naturally emerge in standard macroeconomic models when they are solved and estimated nonlinearly. These features do not require additional frictions, kinks, or time-varying parameters, but are latent properties revealed by fully nonlinear estimation. Our proposed method makes such analysis computationally feasible, lowering the barriers to nonlinear estimation in applied macroeconomics.

Related literature Our contribution to filtering and estimation is a novel framework for the estimation of nonlinear DSGE models. A key benchmark in this literature is Fernández-Villaverde and Rubio-Ramírez (2007), who introduce the particle filter to economics. Particle filtering suffers from high computational costs due to the curse of dimensionality, evident in Gust et al. (2017) who estimate a globally solved model using the particle filter, needing many particles and parallel computing with many nodes, taking six minutes per likelihood evaluation. Similarly, Farmer (2021) proposes a discretization filter for second-order perturbations, but discretizing the state space is computationally intensive and also suffers from the curse of dimensionality. Winschel and Krätzig (2010) aim to resolve this curse of dimensionality by using sparse grid methods and parallelization, but such methods remain computationally costly. In contrast, our proposed method is accurate as well as computationally efficient and feasible on a standard desktop computer without relying on parallelization. Other approaches, such as Borağan Aruoba, Cuba-Borda, and Schorfheide (2018) and Harding, Lindé, and Trabandt (2022), estimate a linearized DSGE model and evaluate its policy implications using a nonlinear model solution. While computationally attractive, this method may be biased when the local perturbation deviates significantly from the global solution – a key issue in our applications given the well-documented nonlinearities in labor search models (see, e.g., Petrosky-Nadeau and Zhang, 2017).

Our quantitative model builds on a long tradition of New Keynesian models and search-and-matching models à la Diamond (1982), Mortensen (1982), and Pissarides (1985), and more recent work integrating both frameworks, such as Christiano, Eichenbaum, and Trabandt (2016) and Thomas (2011), with key contributions from Galí, Smets, and Wouters (2012), Blanchard and Galí (2010), Krause and Lubik (2007), and Gertler, Sala, and Trigari (2008). The model is most similar to Krause, Lopez-Salido, and Lubik (2008) but differs in a key respect: whereas they use log-linearization, which fixes flow profit at its steady-state value, we employ a nonlinear solution that allows flow profit to evolve over time. This enables us to show that monetary transmission varies with labor market conditions, establishing state dependence as a key feature of New Keynesian models with search frictions – our primary contribution relative to this literature.

Our paper relates to a long literature studying the Phillips (1958) curve, including its nonlinearities and time variation in its slope, as examined in Coibion and Gorodnichenko (2015), Smith et al. (2023), and others. Both Smith et al. (2023) (empirically), Benigno and Eggertsson (2023) and Gitti (2024) (model and empirics) argue that the Phillips curve is steeper in tighter labor markets, a finding consistent with our empirical and model-based results. Benigno and Eggertsson (2023) and Gitti (2024) model this within a New Keynesian framework by introducing a kink in wage setting around a market tightness of one, which mechanically induces a kink in the Phillips curve. We differ from this approach by showing that state dependence does not need to be explicitly modeled in a New Keynesian model; rather, it emerges endogenously from its nonlinear solution.

A different but related literature suggests that nonlinearities and state dependence may arise from pricing setting frictions rather than labor market frictions, with contributions from Dotsey, King, and Wolman (1999), Gertler and Leahy (2008), and more recently, Harding et al. (2022) and Harding, Lindé, and Trabandt (2023). Instead, our focus is on labor market frictions as the primary source of state dependence. Finally, a recent study by Lahcen, Baughman, Rabinovich, and van Buggenum (2022) builds on the money search tradition of Berentsen, Menzio, and Wright (2011), showing that small-surplus calibrations imply state-dependent inflation costs. We view the money search literature as complementary, as it abstracts from the nominal rigidities central to New Keynesian frameworks and, consequently, from the role of active monetary policy, which is our primary focus in this application.

Outline The rest of this paper is structured as follows. Section 2 presents the method. Section 3 demonstrates the performance of this method using Monte-Carlo simulations from the Diamond-Mortensen-Pissarides model and the evaluation of the three-equation New-Keynesian model. Section 4 presents the main application. Section 5 presents the quantitative model results and supporting reduced-form evidence, and Section 6 concludes.

2 Local Linear Approximations and Time-Varying Kalman Filtering

In this section, we present the main methodological contribution: a full-information estimation method for DSGE models solved using local linear approximation methods. The key insight of our proposed method is the integration of the solution method within the filtering algorithm, while traditional approaches treat solving and filtering as two independent steps of the estimation approach. We center our discussion on our preferred computational algorithm, Taylor projections (Fernández-Villaverde and Levintal, 2018; Levintal, 2018), as we find it to be computationally more efficient than alternative methods. However, the filtering approach outlined below is more general and can be applied to other local linearization techniques, such as Dynamic Perturbation, as in Mennuni et al. (2024).

2.1 Local Linear Approximation Methods

A standard Dynamic Stochastic General Equilibrium (DSGE) model can be represented as follows (Schmitt-Grohé and Uribe, 2004):

$$\mathbb{E}_t \left[f(x_{t+1}, x_t, z_{t+1}, z_t, \eta_{t+1}) \right] = 0, \tag{1}$$

where x_t represents the control variables, and the state variables z_t can be partitioned into endogenous variables $z_{1,t}$ and exogenous variables $z_{2,t}$. Exogenous variables are assumed to follow a stationary VAR(1): $z_{2,t+1} = Az_{2,t} + B\eta_{t+1}$, where η_t is a vector of Gaussian exogenous shocks. Endogenous variables $z_{1,t+1}$ depend on z_t through an unknown nonlinear mapping. Hence, a solution to (1) is characterized by two implicit functions,

$$z_{t+1} = h(z_t)$$

and the policy function:

$$x_t = g(z_t).$$

A and B are assumed to be known functions of structural parameters (that may have to be estimated), while g and h are the unknown functions that we wish to solve for.

Taylor Projection (Levintal, 2018) The Taylor projection method consists of four main steps. In this exposition and in our filtering method, we focus on the first-order Taylor projection.

In the first step, the policy function $g(z_t)$ and law of motion $h(z_t)$ are approximated using linear functions:

$$\tilde{g}(z_t; \Theta) = \Theta_1 + \Theta_2 z_t \tag{2}$$

$$\tilde{h}(z_t; \Theta) = \Theta_3 + \Theta_4 z_t \tag{3}$$

In step two, the residual function $R(z_t, \Theta)$ is constructed by substituting $x = \tilde{g}(z_t; \Theta)$ and $z_{t+1} = \tilde{h}(z_t; \Theta)$ into the equilibrium condition (1):

$$R(z_t, \Theta) = \mathbb{E}_t \left[f(\tilde{g}(\tilde{h}(z_t)), \tilde{g}(z_t), \tilde{h}(z_t), z_t, \eta_{t+1}) \right] \approx 0. \tag{4}$$

The objective is to find Θ such that the residual function is equal to zero.

To do so, in step three, the residual function is approximated using a Taylor series expansion around a specific point z_0, retaining terms up to first order:

$$R(z_t, \Theta) \approx R(z_0, \Theta) + \left. \frac{\partial R(z, \Theta)}{\partial z} \right|_{z=z_0} (z - z_0). \tag{5}$$

At last, in step four, one solves for the optimal Θ^* such that the Taylor series satisfies:

$$R(z_0, \Theta) = 0$$
$$\frac{\partial}{\partial x} R(z, \Theta)\Big|_{z=z_0} = 0.$$

Since z may consist of multiple state variables, our notation for the first-order derivatives intends to capture all partial derivatives. Solving this system gives a set of state-dependent parameters $\Theta^* = \Theta(z_0)$, yielding an approximate solution $\tilde{g}(z, \Theta^*)$ and $\tilde{h}(z, \Theta^*)$ around z_0. In our examples that follow, we will make more explicit how these steps work.

Dynamic Taylor Projection Instead of computing a Taylor Projection at a single point z_0, one can compute a Taylor Projection at each point along a simulation path, similar in spirit to the Dynamic Perturbation method in Mennuni et al. (2024). This yields highly accurate results, as we show in our simulations below. Given that each solution $\tilde{g}(z, \Theta^*)$ and $\tilde{h}(z, \Theta^*)$ is accurate locally around z_0, as long as variables do not change much from one time period z_t to the next z_{t+1}, the mistakes made by the dynamic implementation of the solution method will be small. Since economic systems, e.g., business cycles, tend to be relatively slow-moving, this is typically the case.

Avoiding Numerical Integration Solving the nonlinear system to recover policy coefficients involves handling expectation operators. Prior studies approximate expectations using monomial integration. We show in the Appendix that, in our applications, assuming local policies are linear in logs, expectations can be solved analytically, yielding an exact nonlinear system. This approach has two advantages: (i) it eliminates errors from integral approximations and (ii) accelerates residual evaluation while providing exact Jacobian terms, speeding up Newton iterations. While not essential, models designed for log-linearization may share this feature, making exact expectation solutions a practical advantage.

2.2 Time-Varying Kalman Filter

Consider a standard linear state space model:

$$y_t = Z s_t + d + \varepsilon_t \tag{6}$$
$$s_{t+1} = T s_t + c + R\eta_t \tag{7}$$

with ε_t i.i.d. measurement error with variance-covariance matrix H, and η_t i.i.d. with variance-covariance Q. y_t is the set of observed variables, s_t are the states, which are assumed unobserved.

It is well-known that linearized DSGE models can be mapped into a linear state space form, and hence, its parameters can be estimated using standard Kalman filtering approaches. For a textbook treatment, consider Herbst and Schorfheide (2016). For a standard model that is log-linearized around the steady state, the matrices Z, d, H, Q, T, c and R depend on structural model parameters, but are constant and do not vary over time.

Our contribution is the insight that when a model is solved using a local linear approximation method such as Levintal (2018)'s Taylor Projection method outlined above, the local linear model solution can be mapped into a *time-varying* state space model, and the model's states can be filtered using a time-varying Kalman Filter, which, in that case, is the optimal linear filter.

To map things back into previous notation, note that the unobserved states s_t of the state-space model will be *known* functions of model state variables z_t, often-times this is a one-to-one mapping. We can generally assume that if one knows s_t, z_t is uniquely recoverable.

Assume the Taylor Projection method around a specific point z is used to obtain an approximation $x_t = \Theta_1^z + \Theta_2^z z_t$ and $z_{t+1} = \Theta_3^z + \Theta_4^z z_t$, where the superscript z indicates that these are the solution matrices based on the Taylor projection around z. Given that our model solution is now fully characterized by linear equations, we can map the model solution into a linear state space form:

$$y_t = Z_z s_t + d_z + \varepsilon_t \tag{8}$$

$$x_{t+1} = T_z s_t + c_z + R\eta_t \tag{9}$$

where Z_z, T_z, d_z and c_z contain elements of the matrices Θ_1^z, Θ_2^z, Θ_3^z, Θ_4^z of the local approximation of the model solution. For the models we consider in this paper, the variance-covariance matrices H and Q, as well as shock selection matrix R do not have state-dependence when using this solution method, so these matrices do not need subscripts z. Innovation matrix η_t corresponds exactly to η_t in our general notation for the DSGE model above.

The Kalman filter is valid and optimal, even when the system matrices Z, d, T and c are time-varying, as long as this time-dependence is fully known when conditioning on the information set $\mathcal{Y}_{t-1} = \{y_{t-1}, y_{t-2}, ...\}$. This follows from the recursive definition of the Kalman filter.[6] This allows us to replace the subscript z in Equations (8)-(9) by $\hat{s}_{t|t-1}$, our prediction for the state s_t using all information up to and including time $t-1$, and implies the filtering procedure outlined in Algorithm 1. Algorithm 2 in Appendix A outlines the Smoothing algorithm.

In words, what Algorithm 1 does is it computes a local linear approximation of the policy functions g and law of motion h at the best prediction for the current state s_t, denoted $\hat{s}_{t|t-1}$ (which implies a prediction for z_t) based on all data up to and including $t-1$. We can map this linear approximate model solution into the linear state space form of Equations (8)-(9), and use the Kalman filter to obtain a prediction for the next state, $\hat{s}_{t+1|t}$. We then repeat the procedure, by computing a local linear model solution around $\hat{s}_{t+1|t}$, mapping this model solution into state space form, and

[6]This is a well-established result; see, for example, Durbin and Koopman (2012) and Hamilton (1994).

Algorithm 1 Time-Varying Kalman Filtering with Taylor Projections

Initialize: Set $t = 0$, start s_0 in the steady state, and initialize covariance $P_{0|0}$.
while $t \leq T$ **do**
Taylor Projection Step:
Solve the model using Taylor Projections at $\hat{s}_{t|t-1}$, obtaining the time-varying system:

$$y_t = Z_{\hat{s}_{t|t-1}} s_t + d_{\hat{s}_{t|t-1}} + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, H) \tag{10}$$

$$x_{t+1} = T_{\hat{s}_{t|t-1}} s_t + c_{\hat{s}_{t|t-1}} + R\eta_t, \quad \eta_t \sim \mathcal{N}(0, Q) \tag{11}$$

Update Step:
Compute the Kalman gain:

$$F_t = Z_{\hat{s}_{t|t-1}} P_{t|t-1} Z'_{\hat{s}_{t|t-1}} + H \tag{12}$$

$$K_t = T_{\hat{s}_{t|t-1}} P_{t|t} Z'_{\hat{s}_{t|t-1}} F_t^{-1} \tag{13}$$

Update the state estimate using the new observation y_t:

$$v_t = y_t - Z_{\hat{s}_{t|t-1}} \hat{s}_{t|t-1} - d_{\hat{s}_{t|t-1}} \tag{14}$$

$$\hat{s}_{t|t} = \hat{s}_{t|t-1} + P_{t|t-1} Z_{\hat{s}_{t|t-1}} F_t^{-1} v_t \tag{15}$$

Update the covariance:

$$P_{t|t} = P_t - P_t Z'_{\hat{s}_{t|t-1}} F_t^{-1} Z_t P_{t|t-1} \tag{16}$$

Prediction step:
Compute the predicted state estimate:

$$\hat{s}_{t+1|t} = T_{\hat{s}_{t|t-1}} \hat{s}_{t|t} + c_{s_{t|t-1}} \tag{17}$$

Compute the predicted covariance:

$$P_{t+1|t} = T_{\hat{s}_{t|t-1}} P_{t|t} T'_{\hat{s}_{t|t-1}} + RQR' \tag{18}$$

Advance time: Set $t \leftarrow t + 1$ and repeat.
end while

obtaining a new prediction $\hat{s}_{t+2|t+1}$, etc. The Kalman Smoother in Algorithm 2 uses the same system matrices as Algorithm 1, so do not require re-solving the model.

2.3 TP+TV-KF versus the Extended Kalman Filter

To better understand our proposed estimation framework, and contrast it to existing frameworks, it is useful to compare the framework to the setting where one would solve the model globally, and use the Extended Kalman Filter (EKF) to approximate the model's likelihood. The distinction between the Extended Kalman Filter (EKF) and our proposed time-varying Kalman filter based on Taylor Projection (TP+TV-KF) is best understood by first considering the structure of a general nonlinear (potentially time-varying) state-space model:

$$y_t = Z(s_t) + \varepsilon_t, \tag{19}$$

$$s_{t+1} = T(s_t) + R\eta_t, \tag{20}$$

where ε_t and η_t are innovations with variance-covariance matrices H and Q, respectively.[7] Following notation in Durbin and Koopman (2012), define

$$\dot{Z}_t = \left.\frac{\partial Z(s_t)}{\partial s_t'}\right|_{s_t=\hat{s}_{t|t-1}}, \quad \text{and} \quad \dot{T}_t = \left.\frac{\partial T(s_t)}{\partial s_t'}\right|_{s_t=\hat{s}_{t|t}}$$

Rather than filtering the nonlinear system (19)-(20) directly, the EKF applies the Kalman filter on the linearized system:

$$y_t = \dot{Z}_t s_t + d_t + \varepsilon_t, \quad d_t = Z(\hat{s}_{t|t-1}) - \dot{Z}_t \hat{s}_{t|t-1}, \quad \varepsilon_t \sim (0, H)$$
$$s_{t+1} = \dot{T}_t s_t + c_t + R\eta_t, \quad c_t = T(\hat{s}_{t|t}) - \dot{T}_t s_{t|t}, \quad \eta_t \sim (0, Q)$$

To understand the distinction between the EKF and TP+TV-KF, first suppose that the global model solution is known, that is, $x_t = g(z_t)$ and $z_{t+1} = h(z_t)$, are the solutions to the equilibrium system (1), they exist and are known. These policy functions map into the nonlinear state-space system (19)–(20) through $Z(s_t)$ and $T(s_t)$.

In the EKF, the first-order Taylor expansion of $Z(\cdot)$ and $T(\cdot)$ effectively linearizes the underlying global policy functions $g(z_t)$ and $h(z_t)$. Consequently, the key difference between the EKF and TP+TV-KF lies in when and how the linearization is performed within the algorithm. A first-order Taylor expansion of an optimal policy rule $x_t = g(z_t)$ around a point z is, in general, no longer an optimal policy and violates equilibrium conditions, except at the non-stochastic steady state. In contrast, the TP solution enforces that the residuals of the equilibrium conditions are exactly zero at the local point of approximation, z_t. This fundamental difference implies that the EKF applied to the true global solution and the TP+TV-KF generally do not coincide.

A deeper distinction concerns the treatment of the composition $g(h(z_t))$. In TP+TV-KF, this composition is determined by solving the full system of equilibrium conditions locally, ensuring that model-consistent dynamics are preserved at each step. In contrast, in the EKF, the coefficients are obtained by a Taylor expansion of the the fixed global policy functions, and thus the resulting linearized dynamics do not generally satisfy first-order optimality conditions. We elaborate on these differences with a simple neoclassical growth model example in Appendix F.

[7] The EKF can also accommodate time-varying volatility depending on s_t, but we omit these extensions here for simplicity.

Practical limitations of global solutions also motivate the use of local approximations. First, computing a global solution for use in the EKF is computationally costly, often requiring several minutes per parameter set (e.g., three minutes in Gust et al. (2017)), and must be repeated for each draw in Bayesian estimation. Second, global solutions are inherently approximate: policy functions satisfy equilibrium conditions only at a finite set of nodes, and accuracy may deteriorate elsewhere. More critically, EKF accuracy depends on the derivatives of the policy functions, which polynomial approximations may capture poorly. In contrast, local linear approximations ensure derivative accuracy by construction, as they directly solve the first-order conditions at each point. Concluding, in practice, our algorithm may offer gains in both computational speed and filtering accuracy.

3 Monte Carlo: Filtering and Estimation of the DMP Model

In this section, we compare the performance of our proposed estimation framework to existing methods using simulations of the Diamond-Mortensen-Pissarides model. This model is known to have significant nonlinear dynamics (Petrosky-Nadeau and Zhang, 2017), making it a natural laboratory for the proposed method. We show that local linear approximation methods lead to highly accurate solutions, and, moreover, full-information estimation of the model with the proposed method achieves the same (or better) accuracy as a global solution coupled with particle filtering at a fraction of the computational cost.

3.1 Model Description

We briefly describe the model environment, as the model is standard. The only non-standard assumption is the one made with respect to the timing of separations and hires. Following the literature on labor search in NK models (Christiano et al., 2016), we assume workers who lose their jobs at the beginning of the period can search within the period for a new job. This assumption is typically justified as a way of making low-frequency (quarterly) NK models conform with labor search where most unemployment spell last only a few weeks.[8]

Firms maximize the present discounted value of profit taking wages as given

$$\mathbb{E}_0\left[\sum_{t=1}^{\infty}\beta^t\left\{(Z_t - W_t)\,N_t - \kappa V_t\right\}\right]$$

subject to

$$N_t = (1-\delta)N_{t-1} + q_t V_t,$$

where the flow payoff is output $Z_t N_t$, net of wage expenditures $W_t N_t$ and vacancy posting costs $\kappa V_t(j)$, δ is the job destruction rate, and q_t the vacancy filling rate. In equilibrium, the Nash wage

[8] According to the estimate of Ahn and Hamilton (2022), two-thirds of workers exit unemployment within the first quarter of an unemployment spell.

is given by $W_t = \eta Z_t + (1-\eta)\nu$, where η is the worker's bargaining weight, and ν is the outside option of the worker. This leads to job creation condition

$$\frac{\kappa}{q_t} = (1-\eta)(Z_t - \nu) + (1-\delta)\beta E_t\left[\frac{\kappa}{q_{t+1}}\right],$$

which is the sole equilibrium condition in the model. We assume productivity is stochastic, and log productivity, denoted z_t, follows an AR(1):

$$z_{t+1} = \rho_z z_t + \varepsilon_{t+1}.$$

3.2 Model Solution

The DMP model above has one state (TFP, z_t) and one policy to solve for: market tightness $\theta_t = g(z_t)$. In the local linear approximation, we approximate the policy function by:

$$\theta_t = \Gamma_1 + \Gamma_2 z_t$$

which implies $\theta_{t+1} = \Gamma_1 + \Gamma_2\rho_z z_t + \Gamma_2\varepsilon_{t+1}$ by iterating the law of motion for z_t one period forward.

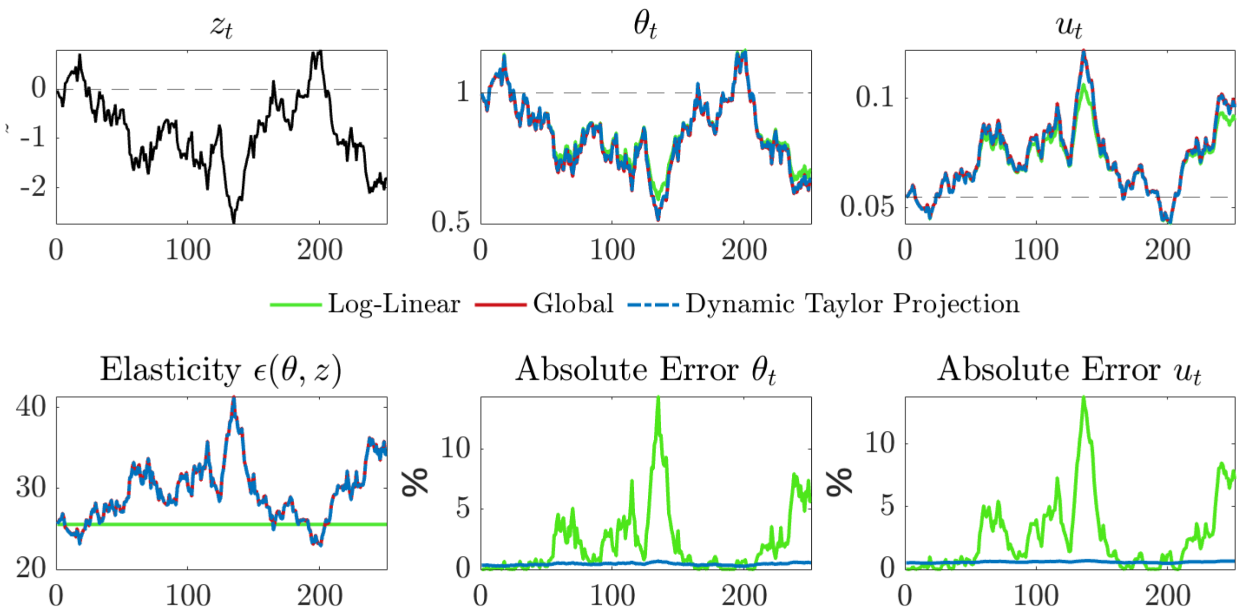
The residual functions for the DMP model, used to solve for the local policy parameters, follow directly from the job creation condition in Equations (3.1), and are given by:

$$\begin{aligned}
R(z,\Gamma)|_{z=z_t} &= \frac{\kappa}{q(\theta_t)} - (1-\eta)(e^{z_t} - b) - \beta(1-\delta)\mathbb{E}_t\left[\frac{\kappa}{q(\theta_{t+1})}\right] \\
\frac{\partial R(z,\Gamma)}{\partial z}|_{z=z_t} &= \frac{-\kappa q'(\theta_t)}{q(\theta_t)^2}\Gamma_2 - (1-\eta)e^{z_t} + \beta(1-\delta)\rho_z\Gamma_2\rho_z\mathbb{E}_t\left[\frac{\kappa q'(\theta_t)}{q(\theta_t)^2}\right]
\end{aligned}$$

This implies two equations in two unknowns Γ_1, Γ_2, which can be solved for numerically.

Figure 1 shows how a global model solution, approximating the unknown policy function by projection onto a set of polynomial basis functions, compared to a dynamic Taylor projection that computes the local linear approximation at each point of the solution path, and how these both compare to a log-linear model solution. The top row shows simulated paths for TFP, which are the same for all three methods, and the implied market tightness and unemployment rate. As can be seen, the simulation from the global model solution and dynamic Taylor projection follow each other closely. The log-linear solution deviates considerably, as measured by the two panels in the bottom row, showing the absolute error compared to the global model solution in market tightness and the unemployment rate, which can be above 10% for the log-linear solution, while for the dynamic Taylor projection, is always low.

Figure 1: Comparison of log-linearization, global model solution, and dynamic Taylor projection for the DMP model.



The first panel in the second row shows that the local linear model solution also accurately tracks a key statistic of the DMP model: time-variability in the elasticity of market tightness and productivity. In a log-linear model solution, this elasticity is constant, since risk does not affect elasticities. Note that even in a second-order perturbation, this elasticity would be constant (Schmitt-Grohé and Uribe, 2004), because uncertainty only shifts the constant in the decision rule in a second order perturbation, and does not affect elasticities. Concluding, local linear approximations are a highly accurate and computationally efficient alternative to global model solutions.

3.3 State-Space Representation

To make explicit how the Taylor projection solution maps into a linear state space model, assume we observe market tightness and wages, with measurement errors. Our local linear model solution around z is given by:

$$\theta_t = \Gamma_{1,z} + \Gamma_{2,z} z_t$$

and $w_t = \log W_t$ follows as:

$$w(z_t) \approx w(z) + \frac{\partial w(z)}{\partial z}(z_t - z)$$

This implies that the state-space representation follows as:

$$\begin{pmatrix} \theta_t^{\text{obs}} \\ w_t^{\text{obs}} \end{pmatrix} = \begin{pmatrix} \Gamma_{1,z} \\ \ln\left(\eta e^z + (1-\eta)b\right) - \frac{z\eta e^z}{\eta e^z + (1-\eta)b} \end{pmatrix} + \begin{bmatrix} \Gamma_{2,z} \\ \frac{\eta e^z}{\eta e^z + (1-\eta)b} \end{bmatrix} z_t + \underbrace{\varepsilon_t}_{\text{meas. error} \sim N(0,Q)}$$

$$z_{t+1} = \rho_z z_t + \eta_t, \quad \eta_t \sim (0, \sigma_z^2).$$

We can then use Algorithm 1 to filter the underlying states and evaluate the likelihood of this model.

3.4 Monte Carlo

In this subsection, we present a simulation experiment based on 1,000 simulated datasets for two different lengths: $T = 200$, which is comparable to the data we use when we estimate the NKDMP model, and $T = 1000$ to provide some insight into asymptotic behavior. The experiment uses the DMP model presented above. For the first set of results, we assume we observe market tightness with measurement error which has a standard error equal to 25% of the standard deviation of the true underlying variable.

The experiment first solves the DMP model using a global solution method and generates simulated data from this solution. Then, it applies four different solution plus filtering approaches:

1. Linear perturbation with a standard Kalman filter (KF).
2. Global solution method with a particle filter (PF).
3. Taylor Projection (TP) with the time-varying Kalman filter (TP+TV-KF) in Algorithm 1.
4. Global solution method with Extended Kalman filter (EKF)

We fix the structural model parameters to their true values (see Table 1 notes), and our goal is to assess how well each method recovers the unobserved states from the simulated data. Table 1 evaluates performance by comparing the R-squared (R^2) of the filtered states with the true states, along with the mean squared error (MSE) and mean absolute error (MAE). Additionally, the table reports computation times for each approach.

Focusing on computation time, for $T = 1000$, Table 1 shows that TP+TV-KF is about three times slower than Linear+KF, while Global+PF is several orders of magnitude slower than TP+TV-KF. Using a global model solution in combination with EKF is about three times as slow as TP+TV-KF for $T = 1000$. In terms of accuracy, for low T, TP+TV-KF consistently outperforms both the global model + PF and the linear model + KF, across all three metrics. In line with our discussion on EKF above, TP+TV-KF and Global+EKF are very comparable in accuracy. For larger T, Global+PF with a sufficient number of particles does improve and approaches the accuracy of TP+TV-KF, but

Table 1: Comparing Solution and Filtering Methods for the DMP model. MSE is scaled by the (unconditional) variance of the variable, MAE by the standard deviation.

	Linear + KF	Global + EKF	Global + PF			TP + TV-KF
Particles:			200	2,000	20,000	
			T = 200			
R^2 Market Tightness	0.959	0.962	0.950	0.951	0.952	0.962
R^2 TFP	0.954	0.961	0.949	0.951	0.951	0.961
MSE Market Tightness	0.040	0.037	0.048	0.047	0.047	0.037
MSE TFP	0.066	0.020	0.026	0.025	0.025	0.020
MAE Market Tightness	0.157	0.154	0.160	0.158	0.158	0.154
MAE TFP	0.176	0.110	0.117	0.114	0.114	0.110
Computation time (sec)	0.001	0.101	0.148	0.564	4.523	0.007
			T = 1,000			
R^2 Market Tightness	0.964	0.968	0.965	0.966	0.967	0.968
R^2 TFP	0.949	0.967	0.965	0.966	0.966	0.967
1e3 × MSE Market Tightness	0.035	0.031	0.033	0.032	0.032	0.031
1e6 × MSE TFP	0.077	0.028	0.030	0.029	0.029	0.028
10 × MAE Market Tightness	0.146	0.140	0.143	0.140	0.140	0.140
1e3 × MAE TFP	0.193	0.133	0.137	0.134	0.133	0.133
Computation time (sec)	0.008	0.109	0.359	2.487	22.62	0.032

Note: Calibration: $r = 0.04$, $\beta = \left(\frac{1}{1+r}\right)^{1/4}$, $\alpha = 0.7$, $\rho_z = 0.985$, σ_z=0.0015, $\eta = 0.5$, $b = 0.94$, $n_{ss} = 0.945$, $q_{ss} = 0.7$, $\theta_{ss} = 1$.

remains dominated. For all sample sizes, Linear+KF performs considerably worse than the other methods. We conclude that the combination of Taylor Projections and the Time-Varying Kalman Filter is an accurate and computationally efficient solution and filtering approach.

3.5 Comparing Posteriors

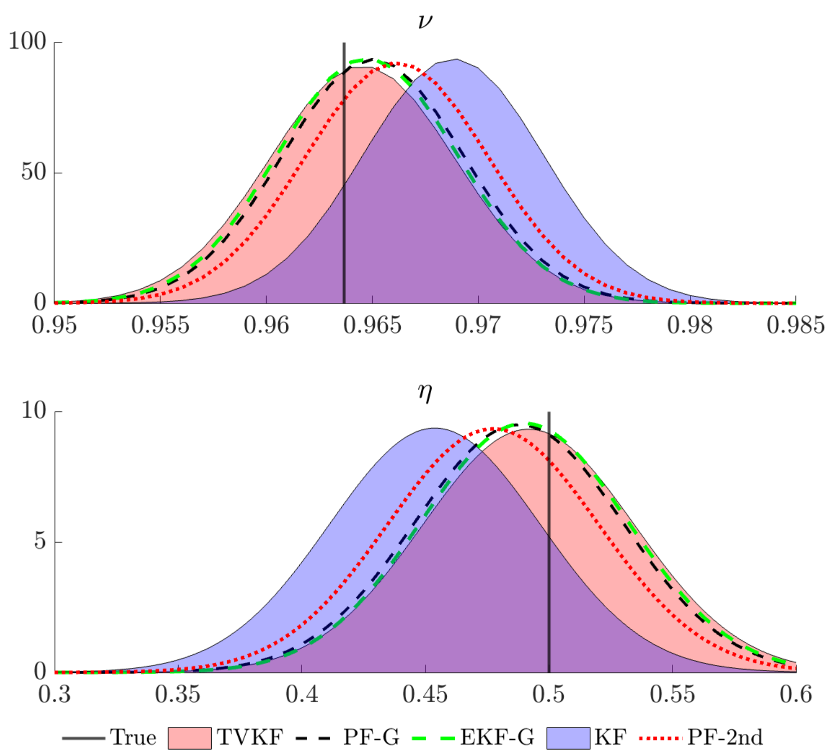
Next, this subsection compares posteriors when the above-described filters are used in a Bayesian estimation routine with a Random Walk Metropolis Hastings algorithm for the posterior draws. We simulate a data set of length $T = 500$ from the global model solution parameterized as in the previous subsection. We assume we observe both market tightness and wages, both with measurement error. For market tightness, the standard deviation of the measurement error equals 30% of the standard deviation of the simulated data series, and for wages 50%. In addition to the four estimation routines above, we also add a second-order perturbation of the DMP model, estimated using the particle filter.[9]

Focusing on economic parameters, Figure 2 shows the posterior distribution for the outside option ν and the bargaining power η. Both have a uniform prior $U[0, 1]$. For both parameters, the Kalman

[9] We use the standard implementation in Dynare.

filter estimates are biased and the posterior distribution is shifted. The posterior distributions obtained from TP+TV-KF, Global+PF and Global+EKF closely align, and are close to the true value. A second-order perturbation filtered with particle filtering (red dotted line) only brings us half way from the biased linearized model parameter estimate to the true parameter, and hence is still biased, albeit less than the linear model.

Figure 2: Comparing posteriors for ν (outside option) and η (bargaining power) for the time-varying Kalman filter of Algorithm 1 (pink area), the Particle Filter + global model solution (black dash), Extended Kalman filter with global model solution (green dash), the linearized model with Kalman filter (blue area), and a second order perturbation with particle filter (red dotted).



4 Revisiting the Three-Equation New Keynesian Model

4.1 Model Description

We closely follow the three-equation model in Gali (2015), and keep the description short, providing additional details in the Appendix on both the model and solution method.

Households Households supply labor, save in nominal bonds, and consume. A representative infinitely lived household maximizes the expected discounted sum of utilities

$$\mathbb{E}_0 \left[\sum_{t=0}^{\infty} \beta^t D_t U\left(C_t, N_t\right) \right]$$

subject to a sequence of budget constraints $P_tC_t + B_t = R_{t-1}B_{t-1} + P_tW_tN_t + T_t$, where C_t is consumption of the final good, B_t is an uncontingent nominal bond, R_{t-1} is the nominal interest rate, W_t is the real wage, and T_t contains dividends from firms net of taxes and transfers from the government. We assume utility has the following form

$$U(C_t, N_t) = \begin{cases} \frac{C_t^{1-\tau}-1}{1-\tau} - \chi \frac{N_t^{1+\eta}}{1+\eta}, \text{ if } \tau \neq 1 \\ \log(C_t) - \chi \frac{N_t^{1+\eta}}{1+\eta}, \text{ if } \tau = 1. \end{cases}$$

$d_t = \log(D_t)$ is a random discount factor shock to the utility of consumption and follows an AR(1):

$$d_{t+1} = \rho_d d_t + \varepsilon_t^d, \;\; \varepsilon_t^d \sim N(0, \sigma_d^2).$$

Firms We assume perfectly competitive final good firms combine a continuum of intermediate goods $j \in [0,1]$. Each intermediate good j is produced by a monopolist with technology $Y_t(j) = A_tN_t(j)$ where A_t is an aggregate technology process common to all firms. These firms choose prices $P_t(j)$ and labor inputs $N_t(j)$ to maximize the present discounted value of profit

$$\mathbb{E}_0\left[Q_{0,t}\left\{\frac{P_t(j)}{P_t}A_tN_t(j) - W_tN_t(j) - \frac{\phi}{2}\left(\frac{P_t(j)}{P_{t-1}(j)} - \Pi\right)^2 Y_t\right\}\right], \quad \text{s.t. } A_tN_t(j) = \left(\frac{P_t(j)}{P_t}\right)^{-1/\gamma} Y_t,$$

where $\frac{\phi}{2}\left(\frac{P_t(j)}{P_{t-1}(j)} - \Pi\right)^2 Y_t$ reflects the Rotemberg price-adjustment cost. $a_t = \log A_t$ follows an AR(1) process. Π is steady state inflation.

Government The government levies lump-sum taxes and runs a balanced budget. The monetary authority follows a Taylor type rule

$$\frac{R_t}{R} = \left(\frac{\Pi_t}{\Pi}\right)^{\phi_\pi}\left(\frac{Y_t}{Y}\right)^{\phi_y} e^{m_t}, \qquad m_{t+1} = \rho_m m_t + \varepsilon_t^m, \;\; \varepsilon_t^m \sim N(0, \sigma_m^2),$$

where m_t is an exogenous monetary policy shock that follows an AR(1).

Equilibrium A minimal state representation consists of states $\mathbb{S} = \{a_t, d_t, m_t\}$ and policies $\{\Pi(\mathbb{S}), N(\mathbb{S})\}$ defined through the equilibrium conditions (substituting market clearing conditions where appropriate)

$$\begin{aligned}
\text{Euler/IS}: \ 1 &= \beta \mathbb{E}_t \left[e^{(\rho_d - 1)d_t + \epsilon^d_{t+1}} \left(\frac{N_{t+1}}{N_t} e^{(\rho_a - 1)a_t + \epsilon^a_{t+1}} \right)^{-\tau} \frac{R_t}{\Pi_{t+1}} \right] \\
\text{Pricing/NK-Phillips}: \ \frac{1}{\gamma} \left(1 - \chi N_t^{\eta + \tau} e^{(\tau - 1)a_t} \right) &= 1 - \phi \left(\Pi_t - \Pi \right) \Pi_t \\
&+ \phi \beta \mathbb{E}_t \left[e^{(\rho_d - 1)d_t + \epsilon^d_{t+1}} \left(\frac{N_{t+1}}{N_t} e^{(\rho_a - 1)a_t + \epsilon^a_{t+1}} \right)^{1-\tau} \left(\Pi_{t+1} - \Pi \right) \Pi_{t+1} \right] \\
\text{Taylor Rule}: \ \frac{R_t}{R} &= \left(\frac{\Pi_t}{\Pi} \right)^{\phi_\pi} \left(\frac{N_t Z_t}{Y} \right)^{\phi_y} e^{m_t}
\end{aligned}$$

More details on the model and equilibrium derivation are provided in the Appendix D.

4.2 Model Solution

We solve the model above using local linear approximations, that is, Taylor Projections (Levintal, 2018) of first-order. Given the minimal state representation stated above, we need to solve for two policy functions:

$$g_\Pi(a_t, d_t, m_t; z) = \alpha_{0,z} + \alpha_{1,z} a_t + \alpha_{2,z} d_t + \alpha_{3,z} m_t$$

and

$$g_N(a_t, d_t, m_t; z) = \beta_{0,z} + \beta_{1,z} a_t + \beta_{2,z} d_t + \beta_{3,z} m_t,$$

where the coefficients $\alpha_0, \alpha_1, \alpha_2, \alpha_3, \beta_0, \beta_1, \beta_2, \beta_3$ are allowed to differ at each point of the state-space where the local approximation is computed, reflected by subscript z. In Appendix D, we show how expectation operators of equilibrium conditions can be solved in closed form.

4.3 State-Space Representation

The states of the model are given by $s_t = (a_t, d_t, m_t)$. The state-transition equation is not time-varying, and follows:

$$\begin{pmatrix} a_{t+1} \\ d_{t+1} \\ m_{t+1} \end{pmatrix} = \begin{pmatrix} \rho_a & 0 & 0 \\ 0 & \rho_d & 0 \\ 0 & 0 & \rho_m \end{pmatrix} \begin{pmatrix} a_t \\ d_t \\ m_t \end{pmatrix} + \begin{pmatrix} \varepsilon^a_{t+1} \\ \varepsilon^d_{t+1} \\ \varepsilon^m_{t+1} \end{pmatrix}, \qquad \begin{pmatrix} \varepsilon^a_t \\ \varepsilon^d_t \\ \varepsilon^m_t \end{pmatrix} \sim N\left(0, \mathrm{diag}(\sigma_a^2, \sigma_d^2, \sigma_m^2)\right)$$

Assume we observe log output y_t^{obs}, log inflation π_t^{obs}, and log interest rates r_t^{obs}. Our observation equation, conditional on having solved the model at state z resulting in policy coefficients $\{\alpha_{i,z}\}_{i=0}^3$ and $\{\beta_{i,z}\}_{i=0}^3$, is given by:

$$\begin{pmatrix} y_t^{\text{obs}} \\ \pi_t^{\text{obs}} \\ r_t^{\text{obs}} \end{pmatrix} = \begin{pmatrix} \beta_{0,z} \\ (\pi_{ss} + \alpha_{0,z}) \\ (r_{ss} + \psi_\pi \alpha_{0,z} + \psi_y \beta_{0,z}) \end{pmatrix} + Z \begin{pmatrix} a_t \\ d_t \\ m_t \end{pmatrix} + \eta_t, \qquad \eta_t \sim N(0, \mathrm{diag}(\sigma_y^{ME}, \sigma_\pi^{ME}, \sigma_r^{ME})^2)$$

$$\text{where } Z = \begin{pmatrix} \beta_{1,z}+1 & \beta_{2,z} & \beta_{3,z} \\ \alpha_{1,z} & \alpha_{2,z} & \alpha_{3,z} \\ (\psi_\pi \alpha_{1,z} + \psi_y(\beta_{1,z}+1)) & (\psi_\pi \alpha_{2,z} + \psi_y \beta_{2,z}) & (\psi_\pi \alpha_{3,z} + \psi_y \beta_{3,z}+1) \end{pmatrix},$$

and σ_y^{ME}, σ_π^{ME} and σ_r^{ME} represent the measurement error standard deviations of log output, inflation and interest rates, respectively. Note that the bottom row in Z reflects how interest rates depend on both output and inflation through the monetary policy rule. This linear state-space representation allows us to solve and filter the model as in Algorithm 1.

4.4 Data

For estimation, we use logged real GDP-per-capita [10], detrended using the HP-filter with a smoothing parameter of 1,600. Inflation is quarterly inflation based on the consumer price index [11] and the nominal rate is the quarterly equivalent of the annual federal funds rate[12]. The data are visualized in Figure 3. We estimate the model on the 1966:Q1-2007:Q4 sample.

4.5 Sampling and Run Time

Sampling We use a Random-Walk Metropolis Hastings sampler, with the proposal variance of the algorithm set to the variance-covariance matrix of the loglinearized posterior draws, scaled to obtain an acceptance rate of 30%. We use the same priors as for the linear model, presented in Table 2. We calibrate a subset of parameters: $\gamma = 0.1$, and $\eta = 5$.

Runtime In Matlab, on a standard laptop without parallelization, it takes 7 minutes and 50 seconds to get 100,000 draws from the nonlinear model, about 0.0047 seconds per draw.[13] This is approximately one-and-a-half times longer than it takes to estimate the linear model in Dynare (Adjemian et al., 2024), which takes five and a half minutes.

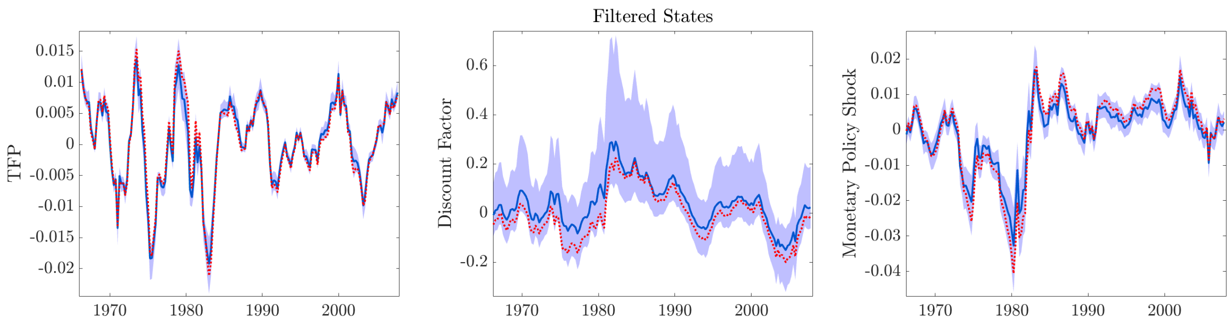
4.6 Estimation Results

Comparing Linear and Nonlinear Estimates Table 2 report the priors – which are common across both the linear model and nonlinear model – and the resulting posteriors for both the linear and nonlinear model estimates. In terms of measurement errors, for the same sample, the nonlinear model has measurement errors for output and interest rates about 30% lower than the linear model, and for inflation, the measurement error is marginally lower as well. For a subset of parameters, the linear and nonlinear model agree closely on the shape and location of the posterior distribution. The biggest differences are in the price adjustment cost parameter ϕ_p which increases from 49 to 76 when estimating the model nonlinearly, and the estimates of the monetary policy

[10] FRED: GDPC1 divided by FRED: CNP16OV
[11] FRED: CPIAUCSL
[12] FRED: DFF
[13] Computations are done on a Apple Macbook Pro with an Apple M3 Max processor and 64GB RAM.

shock process, where the nonlinear model indicates lower persistence of shocks, but with a larger variance.

Figure 3: Filtered States using Nonlinear Model based on Linear Parameter Estimates (red dots) versus Nonlinear Parameter Estimates (blue solid, with 95% CI interval))



Note: Credible intervals only reflects parameter uncertainty, not uncertainty about the state.

To further illustrate the differences between the linear and nonlinear estimates, Figure 3 visualizes the filtered states of both the linear and nonlinear model, and Figure 4 compares impulse response functions. As can be seen, the filtered series for the monetary policy shocks differ because the linear model estimates monetary policy shocks to be more persistent, and also the discount factor shocks differ, although mostly because of a difference in the estimates of the steady state levels of inflation and interest rates, shifting the discount factor down. Figure 4 compares the impulse response functions of the linear and nonlinear model, evaluated at the steady state, which show the implications of different model solutions and parameter estimates.

Figure 4: Impulse Response Functions of Estimated Linear and Nonlinear Model

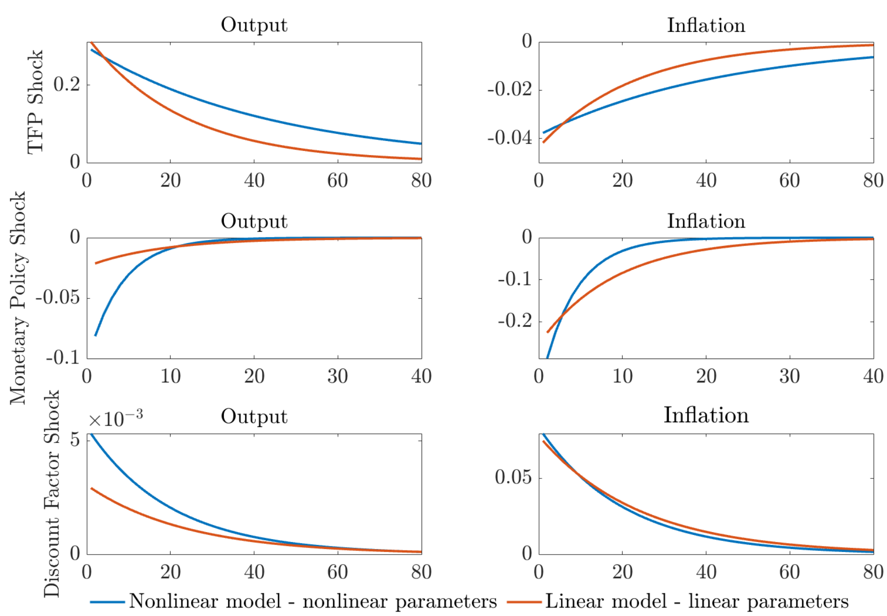


Table 2: *Prior mean and standard deviation, posterior mean and 90% CI*

Parameter	Prior	Linear Posterior		Nonlinear Posterior	
Sample		1966Q1-2007Q4		1966Q1-2007Q4	
	(Mean, Std.)	Mean	[5%, 95%]	Mean	[5%, 95%]
		Economic Parameters			
τ	G (2, 0.5)*	1.51	[1.22, 1.80]	1.68	[1.07, 2.55]
ϕ_p	G(50, 10)*	49.3	[43.5, 56.7]	76.0	[51.5, 101]
ψ_y	G (0.75, 0.25)	0.18	[0.09, 0.27]	0.18	[0.08, 0.30]
ψ_π	G (2.0, 0.25)	2.56	[2.36, 2.92]	2.63	[2.26, 3.06]
		Persistences of shocks			
ρ_m	B (0.7, 0.15)	0.88	[0.80, 0.94]	0.78	[0.68, 0.91]
ρ_d	B (0.7, 0.15)	0.96	[0.94, 0.97]	0.95	[0.92, 0.98]
ρ_a	B (0.7, 0.15)	0.96	[0.92, 0.99]	0.98	[0.95, 1.00]
		Standard deviations of shocks			
$100\times\sigma_m$	IG (0.5, 1)	0.40	[0.31, 0.52]	0.58	[0.34, 0.82]
$10\times\sigma_d$	IG (0.5, 0.5)	0.28	[0.19, 0.45]	0.28	[0.16, 0.48]
$100\times\sigma_a$	IG (1, 1)	0.34	[0.30, 0.38]	0.33	[0.27, 0.39]
		Steady states			
π_{ss}	G (0.01,0.005)	0.012	[0.010, 0.013]	0.011	[0.008, 0.013]
r_{ss}	G (0.016,0.005)	0.016	[0.015, 0.018]	0.014	[0.009, 0.019]
		Measurement error			
$100\times\sigma_y^{ME}$	IG(0.5, 200)	0.12	[0.09, 0.15]	0.08	[0.05, 0.12]
$100\times\sigma_\pi^{ME}$	IG(0.5, 200)	0.43	[0.38, 0.48]	0.41	[0.37, 0.47]
$100\times\sigma_r^{ME}$	IG(0.5, 200)	0.10	[0.08, 0.12]	0.06	[0.04, 0.08]
		Computation time			
Time		5 min 30 s		7 min 50 s	

Note: G is Gamma, IG is inverse Gamma, B is Beta. τ's prior is truncated at 1 from below, and ϕ_p's prior is truncated at 200 from above.

5 A New Keynesian Model with Labor Search

In this section, we present a New Keynesian (NK) model with labor market frictions, following the Diamond-Mortensen-Pissarides (DMP) framework. The model illustrates how the strength of monetary policy transmission depends on firms' flow surplus, leading to state-dependent effects of monetary policy. We provide intuition for this result by analytically log-linearizing the model, showing that surplus scales the sensitivity to shocks. We show how to solve the model using local linear solution methods, how to map the model solution in time-varying state-space form, and consequently how to estimate the model. Using the estimated model and filtered states, we provide both model-based and reduced-form evidence that the Phillips Multiplier, that is, the inflation-unemployment trade-off, is time-varying and decreases during recessions.

5.1 Model

Overview The economy consists of households that consume and supply labor, intermediate good producers that set prices and hire labor in a frictional labor market where workers seek to match with jobs, and final good producers that aggregate intermediate goods while taking prices as given. Wages are determined through Nash bargaining between workers and intermediate goods firms. The government sets the nominal interest rate according to an inertial Taylor rule.

Timing We follow the same timing assumption as in Section 3. Each model period consists of four sub-periods: (i) separations, (ii) search and matching, (iii) bargaining, and (iv) production and consumption/saving.

Households Households are composed of a continuum of members $i \in [0,1]$ who derive utility from consumption $c_t(i)$, and supply one unit of labor $n_t(i) \in \{0,1\}$ if employed and none if unemployed. The members of the household perfectly insure each other against variations in income due to unemployment. Hence, defining $C_t := \int c_t(i)di$, and $N_t := \int n_t(i)di$, households maximize:

$$\mathbb{E}_0\left[\sum_{t=0}^{\infty}\beta^t D_t \frac{C_t^{1-\tau}-1}{1-\tau}\right],$$

subject to the budget constraint

$$P_tC_t + B_t = P_tW_tN_t + P_t(1-N_t)b + R_{t-1}B_{t-1} + T_t,$$

and the law of motion for employment

$$N_t = (1-\delta_t)N_{t-1} + q_tV_t.$$

Here β is the discount factor, D_t is a shock to the value of consumption, $1/\tau$ is the intertemporal elasticity of substitution, P_t is the price of the final good, B_t denotes one-period risk-free bonds with gross return R_t, and T_t denotes lump sum taxes net of profits transferred from firms. W_t are real wages, b denotes the resources provided by the unemployed (e.g., unemployment benefits and home production). The law of motion for employment depends on the exogenous separation rate δ_t and the number of new hires q_tV_t made by firms which the household takes as given, with V_t the number of vacancies and q_t the vacancy filling rate. The first-order condition with respect to consumption implies

$$\lambda_t = D_tC_t^{-\tau},$$

where λ_t is the marginal utility of consumption. We define the stochastic discount factor used by firms as

$$Q_{t,t+1} = \beta \frac{\lambda_{t+1}}{\lambda_t}.$$

The first-order conditions with respect to bonds and consumption imply

$$1 = \beta E_t \left[R_t \frac{D_{t+1}}{D_t} \left(\frac{C_{t+1}}{C_t} \right)^{-\tau} \frac{P_t}{P_{t+1}} \right] = E_t \left[Q_{t,t+1} \frac{R_t}{\Pi_{t+1}} \right]$$

where $Q_{t,t+1}$ is the stochastic discount factor of the households, and $\Pi_t = P_t / P_{t-1}$ Defining the surplus to the household of employment S_t^h using the envelope condition, we obtain

$$S_t^h = W_t - b + E_t \left[Q_{t,t+1}(1 - \delta_{t+1}) S_{t+1}^h \right].$$

which denotes the present value of a marginal employed household member.

Search and Matching The number of aggregate hires is determined by a constant returns to scale matching function $M_t = A_t M(V_t, S_t)$ where A_t is aggregate matching efficiency, V_t is aggregate vacancies and S_t is the number of unmatched workers searching for jobs. Given the timing assumptions, the mass of unmatched workers in the search stage is given by

$$S_t = 1 - N_{t-1} + \delta_t N_{t-1} = 1 - (1 - \delta_t) N_{t-1},$$

that is, the workers who were not matched at the end of the previous production stage plus those who incurred an involuntary separation at the beginning of the period. The probability a vacancy contacts workers can be summarized by the market tightness $\theta_t = V_t / S_t$ as $q_t = M_t / V_t = A_t M\left(1, \theta_t^{-1}\right)$.

Final Goods Producers The finals good producer aggregates intermediate goods $Y_t(j)$, $j \in [0, 1]$, using the technology:

$$Y_t = \left(\int_0^1 Y_t(j)^{1-\gamma} dj \right)^{\frac{1}{1-\gamma}}.$$

The representative firm maximizes profits taking input prices $P_t(j)$ and output prices P_t as given. Profit maximization implies that the demand for input j is given by:

$$Y_t(j) = \left(\frac{P_t(j)}{P_t} \right)^{-\frac{1}{\gamma}} Y_t.$$

Intermediate Goods Producers There is a continuum $j \in [0,1]$ of identical monopolistic intermediate goods producers, with production technology

$$Y_t(j) = Z_t N_t(j)$$

where Z_t is a neutral technology shock, and $N_t(j)$ is the quantity of labor. These firms are monopolists in the product market and hire in frictional labor markets. Price adjustments are subject to a Rotemberg (1982) quadratic adjustment cost

$$AC_t(j) = \frac{\phi}{2}\left(\frac{P_t(j)}{P_{t-1}(j)} - \Pi\right)^2 Y_t,$$

where ϕ is an adjustment cost parameter and Π is steady state inflation. Firms hire labor in a frictional labor market by posting vacancies at a fixed cost c. Hence, the firm's problem is to choose employment $N_t(j)$, vacancies $V_t(j)$, and prices $P_t(j)$ to maximize the present value of real profits discounted by $Q_{0,t}$:

$$\Pi_0(j) = \mathbb{E}_0 \sum_{t=0}^{\infty} Q_{0,t}\left\{\left(\frac{P_t(j)}{P_t}Z_t - W_t\right)N_t(j) - cV_t(j) - AC_t(j)\right\}.$$

subject to

$$Z_t N_t(j) = \left(\frac{P_t(j)}{P_t}\right)^{-\frac{1}{\gamma}} Y_t, \qquad [\omega_t]$$

$$N_t(j) = (1-\delta_t)N_{t-1}(j) + q_t V_t(j). \quad [\mu_t]$$

where the first constraint implies that the choice of output $Y_t(j) = Z_t N_t(j)$ and prices $P_t(j)$ are constrained to lie on the firms' product demand curves. The multiplier on this constraint is given by ω_t. This leads to both an explicit cost of labor the real wage W_t and an implicit cost ω_t as hiring a marginal worker – all else equal – leads to a reduction in prices as firms move down their residual demand curve. The second constraint is the law of motion for employment at the firm and the multiplier μ_t represents the value of a marginal worker to the firm.

Wage Bargaining The wage is determined by the Nash sharing rule

$$\eta\mu_t = (1-\eta)S_t^h.$$

where η is the workers' bargaining weight. This leads to the following expression for the real wage

$$W_t = \eta(1-\omega_t)Z_t + (1-\eta)b$$

where b is the flow value of non-employment and ω_t is the implicit marginal cost of hiring.

Government Monetary policy is determined by a Taylor Rule for the nominal interest rate:

$$\frac{R_t}{R} = \left(\frac{\Pi_t}{\Pi}\right)^{\phi_\pi} \left(\frac{Y_t}{Y}\right)^{\phi_y} e^{m_t}$$

where R_t is the nominal interest rate. The systematic component of monetary policy reacts to the state of the economy, $\Pi_t = P_t/P_{t-1}$ is inflation, Y_t is output, and m_t represents an inertial deviation from the deterministic Taylor Rule. The government levies lump-sum taxes (or provides a subsidy) to run a balanced budget every period.

Government Monetary policy is determined by a Taylor Rule for the nominal interest rate subject to a ZLB constraint:

$$R_t = \max\left\{1, \tilde{R}_t e^{m_t}\right\},$$

where $\tilde{R}_t$ is the systematic component of monetary policy that reacts to the state of the economy, and m_t is the shock. We consider the following Taylor rule:

$$\tilde{R}_t = R\left(\frac{\Pi_t}{\Pi}\right)^{\psi_\pi} \left(\frac{Y_t}{Y}\right)^{\psi_y}$$

The central bank responds to deviations in inflation and output from their targeted values, Π, as well as to the level of output. Note that the nominal interest rate is almost everywhere differentiable, except at the kink where the ZLB begins to bind. The local perturbation method we employ assumes that the equilibrium conditions of the model are differentiable in the state variables. To accommodate the ZLB, we smooth the kink using the softmax function

$$R_t = \zeta \log\left(e^{1/\zeta} + e^{\tilde{R}_t e^{m_t}/\zeta}\right),$$

where the parameter ζ determines how smooth the approximation is and has the property

$$\lim_{\zeta \to 0} R_t = \max\left\{1, \tilde{R}_t e^{m_t}\right\}.$$

Exogenous States

$$\begin{aligned}
z_t &= \rho_z z_{t-1} + \sigma_z \epsilon_t^z \\
d_t &= \rho_d d_{t-1} + \sigma_d \epsilon_t^d \\
m_t &= \rho_m m_{t-1} + \sigma_m \epsilon_t^m \\
s_t &= \rho_s s_{t-1} + \sigma_s \epsilon_t^s \\
a_t &= \rho_a a_{t-1} + \sigma_a \epsilon_t^a
\end{aligned}$$

The discount factor $d_t = \ln D_t$, TFP $z_t = \ln Z_t$, and the monetary shock m_t are assumed to be AR(1) in logs. The separation rate $\delta_t = \delta e^{s_t}$ and matching efficiency $A_t = Ae^{a_t}$ are assumed to have stochastic components that are AR(1) in logs as well. A minimum state variable representation of the model has six states $\mathbb{S}_t = \{N_{t-1}, z_t, d_t, m_t, s_t, a_t\}$ and three policies $\{\Pi(\mathbb{S}_t), \omega(\mathbb{S}_t), \theta(\mathbb{S}_t)\}$.

Equilibrium Conditions We assume vacancy costs and adjustment costs are paid in lump-sum fashion to households, so market clearing implies

$$C_t = Y_t = Z_t N_t.$$

This leads to the following Euler equation

$$1 = \beta \mathbb{E}_t \left[\frac{D_{t+1}}{D_t} \left(\frac{Y_{t+1}}{Y_t} \right)^{-\tau} \frac{R_t}{\Pi_{t+1}} \right].$$

In a symmetric equilibrium all firms set the same prices $P_t = P_t(j)$ and choose the same level of employment $N_t(j) = N_t$ and vacancies $V_t(j) = V_t$ leading to the following pricing equation

$$\frac{\omega_t}{\gamma} = 1 - \phi(\Pi_t - \Pi)\Pi_t + \phi\beta\mathbb{E}_t \left[\frac{D_{t+1}}{D_t} \left(\frac{Y_{t+1}}{Y_t} \right)^{1-\tau} (\Pi_{t+1} - \Pi)\Pi_{t+1} \right].$$

Similarly, optimal vacancy posting and the expression for the Nash wage leads to the following job-creation condition

$$\frac{c}{q_t} = (1-\eta)\left[(1-\omega_t)Z_t - b\right] + \beta\mathbb{E}_t \left[(1-\delta_{t+1}) \frac{D_{t+1}}{D_t} \left(\frac{Y_{t+1}}{Y_t} \right)^{-\tau} \frac{c}{q_{t+1}} \right]$$

The Taylor Rule for the nominal rate and the law of motion for employment close the model.

5.2 State-Dependence

We use the system of log-linearized conditions to think about the deterministic response of the system to a one-time monetary shock $\epsilon_0^m = 1$, and $\epsilon_t^m = 0 \ \forall\, t > 0$, that is, the perfect foresight path. Details on the equilibrium conditions and the derivation of the log-linear system are provided in Appendix C. In this framework, total factor productivity is at steady state, hence $\hat{z}_t = 0$, where the $\hat{\cdot}$ notation denotes a variable in log-deviation from the steady state. This gives us a reduced system of equations (where we omit expectation operators as this is the perfect foresight response):

$$\begin{aligned}
\text{Euler/IS: } \hat{n}_{t+1} &= \hat{n}_t + \frac{1}{\tau}\left[\hat{r}_t - \hat{\pi}_{t+1}\right] \\
\text{Pricing/NK-Phillips: } \hat{\pi}_t &= \beta\hat{\pi}_{t+1} - \frac{1}{\phi}\hat{\omega}_t \\
\text{Job-Creation: } \hat{q}_t &= \frac{\gamma(1-\beta(1-\delta))}{(1-\eta)(1-\gamma-b)}\hat{\omega}_t + (1-\delta)\beta\left[\hat{q}_{t+1} + \hat{r}_t - \hat{\pi}_{t+1}\right] \\
\text{Taylor Rule: } \hat{r}_t &= (1-\rho_R)\left[\psi_1\hat{\pi}_t + \psi_2(\hat{n}_t - \hat{n}_{t-1})\right] + \rho_R\hat{r}_{t-1} + \epsilon_{R,t} \\
\text{LOM: } \hat{n}_t &= (1-\delta)\hat{n}_{t-1} + \delta\left(\hat{q}_t + \hat{v}_t\right)
\end{aligned}$$

As evident from the log-linearized job creation condition, the sensitivity of q_t to ω_t scales inversely with $(1-\eta)(1-\gamma-\nu)$, which represents the steady-state flow profit of the firm. This is closely related to the *fundamental surplus* concept in Ljungqvist and Sargent (2017). When the flow profit is small, shocks propagate more strongly to the vacancy-filling rate q_t, and vice versa. Since q_t determines market tightness, the same mechanism governs the propagation of shocks to market tightness and, consequently, employment and output. Given that inflation depends on $\hat{\omega}_t$ through the New Keynesian Phillips curve, this mechanism also drives state dependence in inflation. The strength of inflation's response to a monetary shock is influenced by the degree of price stickiness ϕ, as specified in the Rotemberg (1982) adjustment cost framework.

In a log-linearized NK-DMP model, the propagation mechanisms depend on steady-state profit $(1-\eta)(1-\gamma-\nu)$ for real variables and on price stickiness ϕ for inflation. However, these elasticities remain constant across different economic states. To accurately capture state dependence, we require a solution method that allows these elasticities to vary with the state of the economy. As we will demonstrate in Section 6, the nonlinear model solution exhibits significant state dependence.

5.3 Solution and Estimation

Here we describe the model solution and mapping to observables in the state space model. First, we define states and policies in terms of log-deviations from their steady state values, e.g. $\theta_t = \ln(\Theta_t/\Theta)$ and $z_t = \ln(Z_t/Z)$. Next, we define the equilibrium conditions in terms of these policies under the assumption that employment n_t, inflation π_t, the marginal cost ω_t and market tightness are linear in the states $x_t = \{n_{t-1}, z_t, d_t, m_t, s_t, a_t\}$. This gives policies of the form.

$$\begin{aligned}
\theta_t &= b_{0,t}^{\theta} + B_t^{\theta} x_t \\
\pi_t &= b_{0,t}^{\pi} + B_t^{\pi} x_t \\
\omega_t &= b_{0,t}^{\omega} + B_t^{\omega} x_t \\
n_t &= b_{0,t}^{n} + B_t^{n} x_t
\end{aligned}$$

Where each B_t^i term is a 1×6 vector of coefficients. Using these local linear policies, the expectation terms of the equilibrium system can be solved in closed form, and the coefficients can be recovered using the Taylor Projection method described in the previous section. In the absence of a binding ZLB constraint, the interest rate can be expressed as a function of the policy functions and model states:

$$r_t = \psi_1 \pi_t + \psi_2 (n_t + z_t) + m_t,$$

which, using the previously defined policy functions, corresponds to a linear projection onto the latent states x_t, making the incorporation of observable nominal interest rates into a linear state-space model straightforward. However, to accommodate periods when the ZLB binds, we smooth the shadow rate using a softmax function. This introduces an additional nonlinear equation. To maintain a linear state-space representation, we approximate the interest rate using a first-order Taylor expansion.

$$r_t = b_{0,t}^r + B_t^r x_t.$$

At states where the interest rate is far from the ZLB, the linear approximation closely matches the unconstrained form. However, as the interest rate approaches the ZLB, the dynamics begin to differ. Additional details and derivations of the local linear solution are provided in Appendix E.

With the local solution to the equilibrium system in hand, we define the state-space mapping between observables and latent states. We consider five observable variables: market tightness $\tilde{\theta}_t$, job-finding rate $\tilde{f}_t$, employment $\tilde{n}_t$, Federal Funds rate $\tilde{r}_t$, and inflation $\tilde{\pi}_t$, where the tildes indicate that these observables may be measured with error. The mapping between these observables and the model variables incorporates measurement errors and is given by:

$$\begin{aligned}
\tilde{\pi}_t &= \pi + \pi_t + \sigma_\pi^{me} \epsilon_t^\pi \\
\tilde{n}_t &= n_t + \sigma_n^{me} \epsilon_t^n \\
\tilde{\theta}_t &= \theta_t + \sigma_\theta^{me} \epsilon_t^\theta \\
\tilde{r}_t &= r + r_t + \sigma_r^{me} \epsilon_t^r \\
\tilde{f}_t &= f_t + \sigma_f^{me} \epsilon_t^f = a_t + (1-\alpha)\theta_t + \sigma_f^{me} \epsilon_t^f
\end{aligned}$$

where π is the natural log of steady-state inflation and r is the log of the steady-state nominal rate. Letting $y_t = \left(\tilde{\pi}_t, \tilde{n}_t, \tilde{\theta}_t, \tilde{r}_t, \tilde{\pi}_t`\right)'$, the state space for of the model at time t is given by

$$\begin{aligned}
y_t &= d_t + Z_t x_t + \varepsilon_t^{me}, \quad \varepsilon_t^{me} \sim N(0, Q) \\
x_t &= c_t + T_t x_{t-1} + R \eta_t^x, \quad \eta_t \sim N(0, H)
\end{aligned}$$

where the measurement equation matrices are

$$d_t = \begin{pmatrix} \pi + b^{\pi}_{0,t} \\ 0 \\ b^{\theta}_{0,t} \\ r + b^{r}_{0,t} \\ (1-\alpha)b^{\theta}_{0,t} \end{pmatrix} \quad Z_t = \begin{pmatrix} b^{\pi}_{1,t} & b^{\pi}_{2,t} & b^{\pi}_{3,t} & b^{\pi}_{4,t} & b^{\pi}_{5,t} & b^{\pi}_{6,t} \\ 1 & 0 & 0 & 0 & 0 & 0 \\ b^{\theta}_{1,t} & b^{\theta}_{2,t} & b^{\theta}_{3,t} & b^{\theta}_{4,t} & b^{\theta}_{5,t} & b^{\theta}_{6,t} \\ b^{r}_{1,t} & b^{r}_{2,t} & b^{r}_{3,t} & b^{r}_{4,t} & b^{r}_{5,t} & b^{r}_{6,t} \\ (1-\alpha)b^{\theta}_{1,t} & (1-\alpha)b^{\theta}_{2,t} & (1-\alpha)b^{\theta}_{3,t} & (1-\alpha)b^{\theta}_{4,t} & (1-\alpha)b^{\theta}_{5,t} & (1-\alpha)b^{\theta}_{6,t}+1 \end{pmatrix}$$

and the state equation matrices are

$$c_t = \begin{pmatrix} b^{n}_{0,t} \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} \qquad T_t = \begin{pmatrix} b^{n}_{1,t} & b^{n}_{2,t} & b^{n}_{3,t} & b^{n}_{4,t} & b^{n}_{5,t} & b^{n}_{6,t} \\ 0 & \rho_z & 0 & 0 & 0 & 0 \\ 0 & 0 & \rho_d & 0 & 0 & 0 \\ 0 & 0 & 0 & \rho_m & 0 & 0 \\ 0 & 0 & 0 & 0 & \rho_s & 0 \\ 0 & 0 & 0 & 0 & 0 & \rho_a \end{pmatrix}$$

We estimate the model on the sample period from 1966:Q1-2019:Q4. The data used for estimation are summarized in Table B2 in the Appendix. This table also details the data transformations applied.

5.4 Details on Empirical Framework: State-Dependent Local Projections

Following Cloyne et al. (2023), one can extend the standard Local Projection (Jordà, 2005) framework to allow for state-dependent impulse responses. Specifically, consider a *state variable* x_t. The state-dependent local projection framework is given by:

$$y_{t+h} = \alpha_h + \beta_h s_t + \beta^x_h s_t(x_t - \bar{x}) + w_t\gamma_h + \varepsilon_t, \quad \text{for } h = 0, \dots, H \text{ and } t = 1, \dots, T, \tag{21}$$

where w_t is a set of control variables, augmented with $(x_t - \bar{x})$, representing deviations of the state variable from its steady-state level $\bar{x}$. Relative to a standard Local Projection, the state-dependent specification introduces the term $\beta^x_h s_t(x_t - \bar{x})$, which captures the interaction between the shock s_t and the state variable x_t. The state-dependent impulse response function is then given by the sum of the direct effect of the shock, $\hat{\beta}_h$, and the state-dependent effect, $\hat{\beta}^x_h(x_t - \bar{x})$.[14] We use an instrumental variable (IV) approach, where the shock s_t is replaced by a policy variable, such as the Federal Funds Rate (FFR), instrumented in a first stage using an exogenous shock series. Given the model includes an interaction term, the instrument set is extended by the interaction between the exogenous shock and the deviation of the state variable from its steady-state level.

Outcome Variables The empirical analysis focuses on the effects of monetary policy shocks on the unemployment rate, vacancies (from the Job Openings and Labor Turnover Survey, and measured

[14] To aid inference, we impose the restriction that the response for each month in a quarter is the same, reducing the number of parameters to be estimated and shortening confidence intervals.

using the Help-Wanted Index from Barnichon (2010) before December 2001), market tightness (the ratio of vacancies to unemployment) and the Consumer Price Index (CPI). Detailed descriptions of the data sources and transformations are provided in Appendix B.

Controls In the local projection instrumental variable framework, we follow the approach of Ramey (2016) by including controls for the CPI, the unemployment rate, the Producer Price Index (PPI), and lags of the monetary policy variable and the dependent variable. Following the recommendations of Bauer and Swanson (2023), the LP-IV framework includes 12 monthly lags.

State Proxies The empirical analysis in this section uses two proxies likely correlated with firm surplus to provide model-free reduced-form evidence on state-dependent monetary policy transmission:

1. Corporate Profits: The value of job creation corresponds to the present value of profits obtained from a marginal hire. While this present value is not directly observable in the data, corporate profits are. In the data, corporate profits are measured as the ratio of profits to the sum of profits and compensation, as reported in the National Income and Product Accounts (NIPA), Table 1.10.

2. Unemployment Rate: This is consistent with model predictions which would indicate that unemployment tracks flow surplus, albeit with a potential lag.

We acknowledge that a causal interpretation of this framework relies on the exogeneity of the monetary policy shocks with respect to both state and outcome variables at all horizons (Gonçalves, Herrera, Kilian, and Pesavento, 2023) — a condition that may not be satisfied. Our aim in this section is more modest: to document correlations indicative of state dependence that are consistent with the theoretical model.

Instruments and Policy Variables We employ three commonly used instruments for the monetary policy rate (which we replace by the one-year U.S. Treasury yield):

1. The narrative monetary policy shock series from Romer and Romer (2004), extended by Wieland and Yang (2020) to 2007;

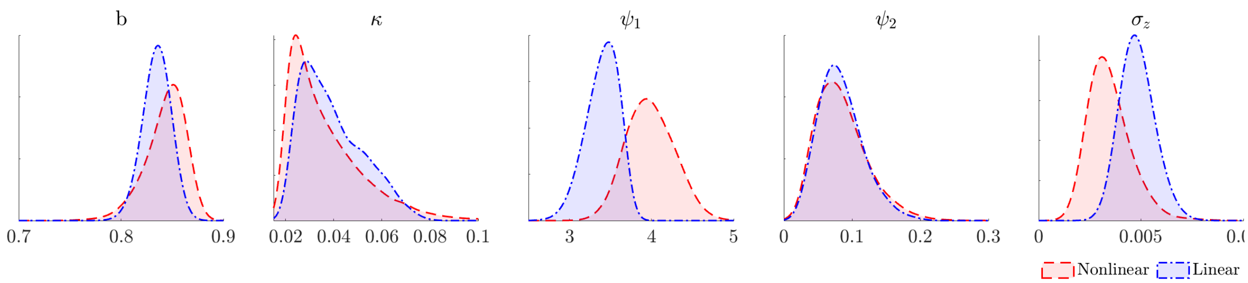
2. The language-processing approach of Aruoba and Drechsel (2024), spanning 1982-2008;

3. The high-frequency approach of Bauer and Swanson (2023), spanning 1988-2019.

6 Model Results

We now present results from the quantitative model, solved using Taylor projections and using the Time-Varying Kalman Filter to infer the underlying states of the economy. This approach enables

Figure 5: Selected Posteriors: linear solution with Kalman filter versus and nonlinear solution (Taylor Projection) with Time-Varying Kalman Filter



us to compute implied impulse response functions over time, as well as other objects of interest to policymakers, and to quantify state-dependence in monetary policy transmission.

6.1 Parameter Estimates

A subset of parameters is calibrated, as reported in Table 3. Priors and posteriors are summarized in Table 4, comparing estimates from the linear and nonlinear models. The estimates differ in several respects. First, the standard deviation of total factor productivity (TFP) shocks is larger in the linear model, reflecting weaker endogenous propagation. Second, the Taylor rule places greater weight on inflation in the nonlinear model. The linear model is estimated on a sample ending in 2007 to avoid the period when the zero lower bound (ZLB) binds, which may further contribute to the differences.

The outside option of workers, b, determines the size of the surplus in the model. To rationalize observed unemployment fluctuations, b must be large relative to the joint surplus of workers and firms. We estimate b by introducing an auxiliary variable $\nu \in [0,1]$, where

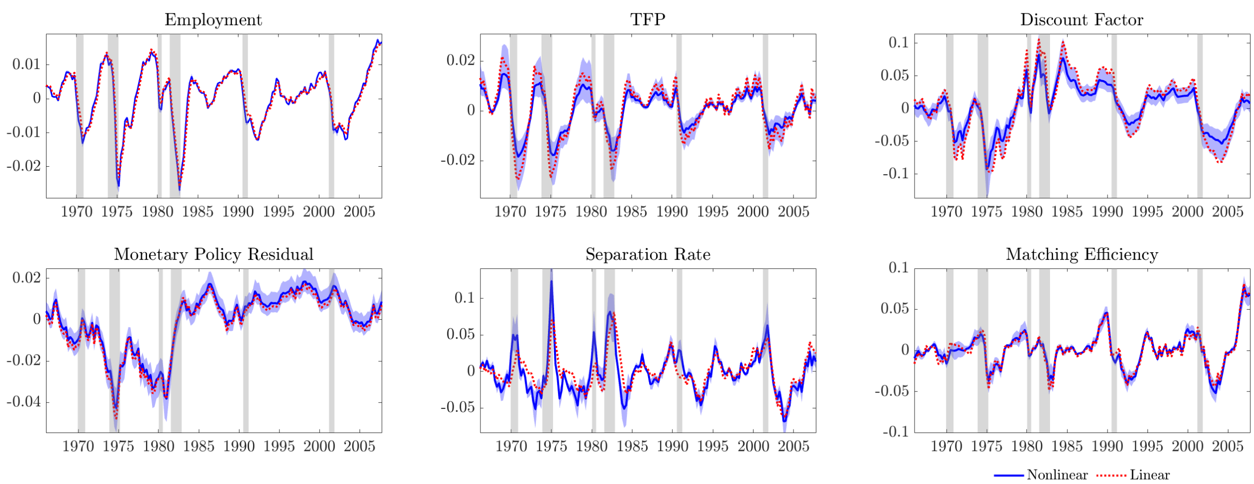
$$b = \nu(1-\gamma)e^{-3\sigma_z/\sqrt{1-\rho_z^2}}.$$

The parameter ν maps b to a fraction of the joint flow surplus corresponding to a productivity realization at the 0.1 percentile of the ergodic distribution. This construction ensures that, conditional on draws of σ_z and ρ_z, the expected surplus remains positive, guaranteeing an interior solution to the job-creation condition (i.e., positive vacancies). Because b is rescaled as a function of σ_z and ρ_z, its posterior distribution differs across models even though the posterior distributions of ν are similar. A selected subset of these posteriors is shown in Figure 5. Differences in estimation and solution methods generate differences in the filtered states, shown in Figure 6. Filtered TFP is substantially more volatile in the linear model than in the nonlinear model, consistent with the parameter estimates. The same holds for discount factor shocks.

Table 3: Calibrated parameters

Parameter	Calibrated value
η: bargaining weight	0.5
τ: risk aversion	2
γ: inverse elasticity of substitution CES aggregator	0.1
q_{ss}: steady state vacancy-filling rate	0.7
θ_{ss}: steady state market tightness	1
n_{ss}: steady state employment	0.945
δ: exogenous job destruction rate	$\frac{q_{ss}\theta_{ss}(1-n_{ss})}{(1-q_{ss}\theta_{ss})n_{ss}}$
s_{ss}: steady state separation rate	$\log \delta$
a_{ss}: steady state match efficiency	$\log q_{ss}$
c: vacancy cost	$\frac{(1-\eta)(1-\gamma-b)q_{ss}}{1-\beta(1-\delta)}$

Figure 6: Filtered States for Linear and Nonlinear Model Solution



6.2 Time-Varying Impulse Responses

An attractive feature of the methodology proposed in this paper is that it permits filtering of the underlying states of the economy and computing the implied impulse response functions at each point in time. The extent to which shock propagation varies over time can be understood through the local linear policy coefficient $\Gamma(x_t)$. In a truly linear model, these coefficients are constant, and impulse responses are invariant to the state. In our setting, this is not the case.

Figure 7 plots the policy coefficients, which can be interpreted as elasticities since all variables are expressed in log-deviations from steady state values, along with their 68% credible intervals, reflecting parameter uncertainty. The results show that the coefficients vary systematically with the state. Elasticities are typically larger in absolute value during recessions, when the model-implied surplus is low, consistent with theoretical predictions. For instance, the effect of monetary shocks

Table 4: Prior distribution with mean and standard deviation, posterior means and standard deviations for log-linear and TP+TV-KF (based on 100,000 draws, acceptance rate of ≈34%, burn-in 20,000, thin 3).

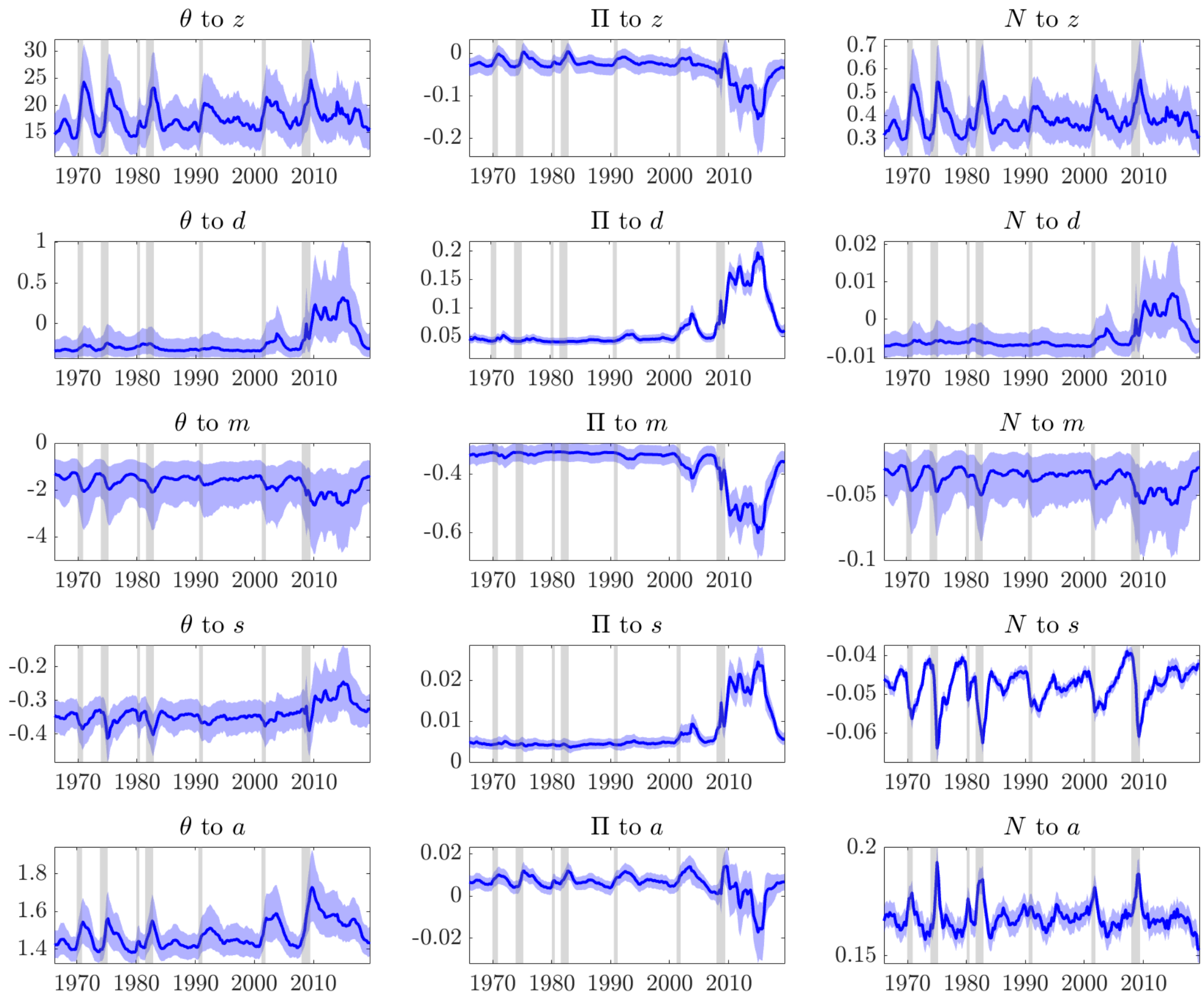
Parameter	Prior Distr. (Mean, Std.)	Posterior Log-linear Mean	Posterior Log-linear Std.	Posterior TP TV-KF Mean	Posterior TP TV-KF Std.
Economic parameters					
ν: flow value unemployment	Beta(0.9, 0.025)*	0.971	0.006	0.976	0.006
κ: slope Phillips curve	Beta(0.05, 0.02)*	0.039	0.013	0.037	0.015
α: matching function	Beta(0.5, 0.1)	0.816	0.023	0.837	0.025
ψ_1: inflation Taylor rule	Gamma(1.5, 0.25)	3.401	0.181	3.985	0.281
ψ_2: output Taylor rule	Gamma(0.25, 0.1)	0.084	0.032	0.085	0.037
Exogenous process parameters					
σ_z: std. TFP shocks	InvGamma(0.02, 0.05)	0.005	0.001	0.004	0.001
σ_d: std. DF shocks	InvGamma(0.05, 0.1)	0.017	0.003	0.013	0.002
σ_s: std. separation shocks	InvGamma(0.05, 0.1)	0.021	0.004	0.042	0.004
σ_m: std. MP shocks	InvGamma(0.02, 0.05)	0.005	0.001	0.005	0.001
σ_a: std. matching efficiency shocks	InvGamma(0.02, 0.05)	0.011	0.001	0.012	0.001
ρ_z: persistence TFP	Beta(0.5, 0.1)	0.949	0.004	0.964	0.010
ρ_d: persistence DF	Beta(0.5, 0.1)	0.890	0.020	0.874	0.018
ρ_s: persistence separations	Beta(0.5, 0.1)	0.770	0.059	0.712	0.038
ρ_r: inertia Taylor rule	Beta(0.5, 0.1)	0.883	0.025	0.886	0.023
ρ_a: persistence matching efficiency	Beta(0.5, 0.1)	0.885	0.028	0.909	0.020
Measurement error					
σ_π^{ME}: ME inflation	InvGamma(0.005, 0.05)	0.0018	0.0002	0.0018	0.0001
σ_N^{ME}: ME employment	InvGamma(0.005, 0.05)	0.0008	0.0001	0.0006	0.0001
σ_θ^{ME}: ME tightness	InvGamma(0.015, 0.05)	0.0152	0.0057	0.0368	0.0045
σ_r^{ME}: ME FFR	InvGamma(0.005, 0.05)	0.0010	0.0001	0.0006	0.0001
σ_s^{ME}: ME separations	InvGamma(0.015, 0.05)	0.0158	0.0018	0.0179	0.0015
Steady states					
π^{ss}: steady state inflation	InvGamma(0.008,0.02)	0.0087	0.0015	0.0097	0.0010
$r_s s$: steady state interest rate	InvGamma(0.013, 0.02)	0.0129	0.0022	0.0138	0.0014

Note: (*) The prior for ν is truncated from above at 0.985 and for κ truncated from below at 0.02.

on market tightness is larger when unemployment is high, and once the ZLB binds, the effects of non-monetary shocks on inflation become large as monetary policy becomes constrained.

To further illustrate the extent of time variation, Figure 8 shows the impulse response functions for unemployment, market tightness, and vacancies at two different dates, computed from the filtered states. The first date, 1973Q2, corresponds to a high-surplus period in which monetary policy shocks propagate little. The second date, 2008Q2, represents a low-surplus period with stronger propagation. In addition to the model-implied impulse responses, we also report results from three different state-dependent local projections, introduced in Section 5.4. The model's predictions for unemployment responses are broadly consistent with the local projections, both in magnitude and

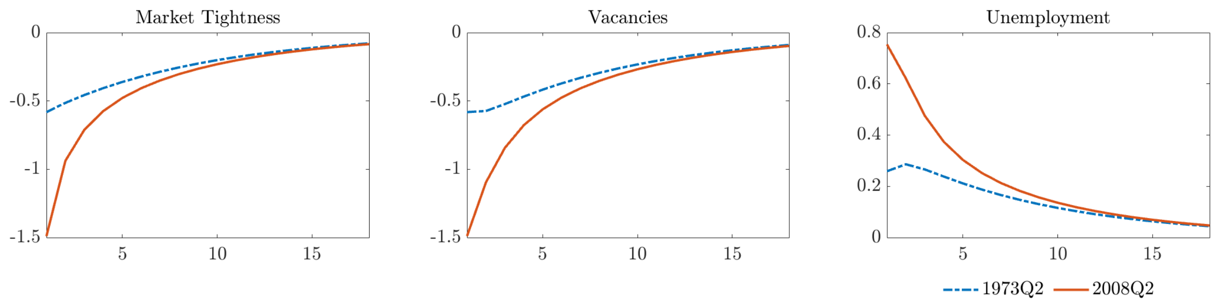
Figure 7: State-Dependent Policy Coefficients, computed at filtered states



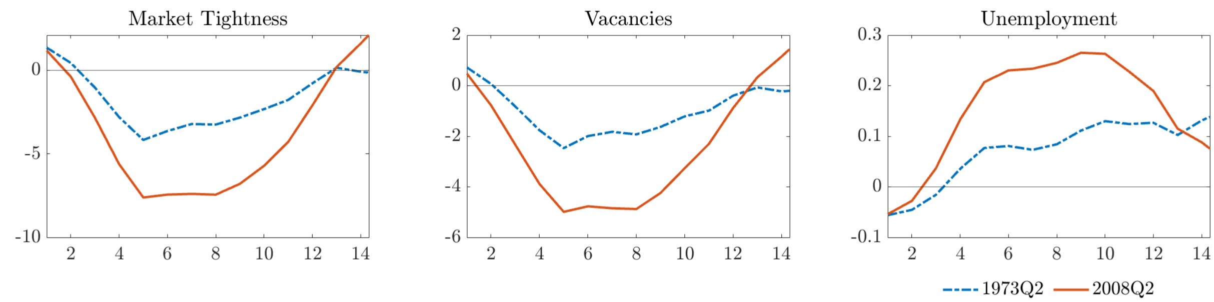
in the extent of state dependence. While the model does not generate hump-shaped responses—an outcome that would require introducing habit formation or additional frictions—this limitation is not central to the analysis. The data show somewhat stronger responses in market tightness and vacancies than the model predicts, but the degree of state dependence is similar.

Figure 8: Impulse Response Functions over Time

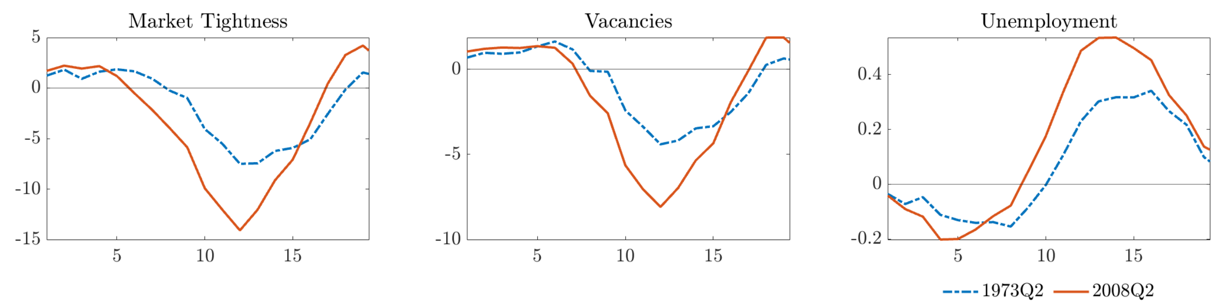
(a) Nonlinear model



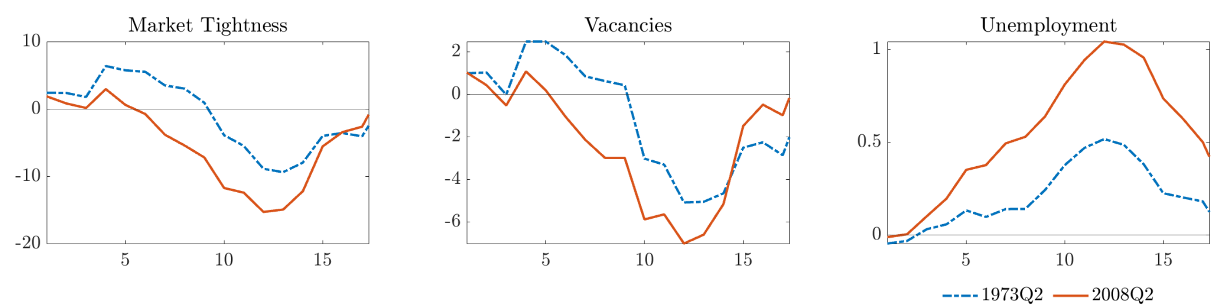
(b) State-Dependent Local Projections with Romer and Romer (2004) shocks and Corporate Profit Proxy



(c) State-Dependent Local Projections with Aruoba and Drechsel (2024) shocks and Corporate Profit Proxy



(d) State-Dependent Local Projections with Bauer and Swanson (2023) shocks and Corporate Profit Proxy



6.3 Phillips Multiplier

A key concern for policymakers is the unemployment-inflation trade-off, which captures the nominal effects relative to the real effects of a monetary shock. However, state dependence in impulse responses alone does not necessarily imply variation in the unemployment-inflation trade-off; this depends on whether the real versus nominal effects differ in relative magnitude across states. To illustrate that the unemployment-inflation trade-off is state dependent, we use the concept of the Phillips multiplier, as in Barnichon and Mesters (2021), motivated by Mankiw (2001). The Phillips multiplier quantifies this trade-off by measuring the average change in inflation caused by an interest rate change that reduces the unemployment rate by one percentage point over h periods:

$$T_h = \frac{\partial \bar{\Pi}_{t:t+h}}{\partial R_t}\Big|_{\varepsilon_t} \Big/ \frac{\partial \bar{U}_{t:t+h}}{\partial R_t}\Big|_{\varepsilon_t}$$

Here $\bar{\Pi}_{t:t+h}$ is the average over variable Π over h time periods, and ε_t is the exogenous change in the interest rates. The statistical counterpart of T_h is the so-called Phillips multiplier (Barnichon and Mesters, 2021):

$$\mathcal{P}_h = \mathcal{R}_h^{\bar{\Pi}} / \mathcal{R}_h^{\bar{U}}, \quad h = 0, 1, 2, \ldots,$$

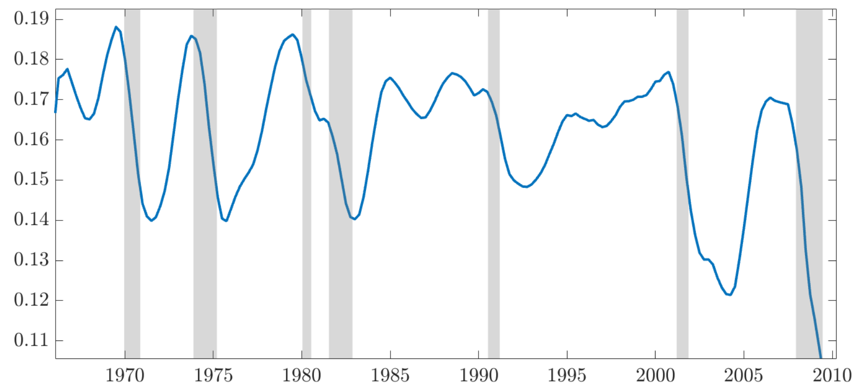
that is, the ratio of the cumulative impulse response function (up to horizon h) of inflation over unemployment. The slope of the Phillips curve and the Phillips Multiplier are strongly connected, but not one-to-one. In the case of state-dependent impulse response functions, the Phillips Multiplier is also state-dependent when the magnitude of the effect of a monetary shock on inflation relative to unemployment depends on the state of the economy.

Using the model to filter the latent states, we compute the implied impulse response of a monetary policy shock at each point in time. These impulse responses, evaluated at each filtered state, allow us to compute the Phillips Multiplier, after which the cumulative impulse responses stabilize as the effects dissipate. We visualize this filtered series in Figure 9 until 2010, because this overlaps with our local projection sample, and for the full estimation sample in Figure 11. As shown, the Phillips Multiplier is generally countercyclical, and, in absolute values, falls during recessions. Several periods stand out where the Phillips curve was particularly flat, typically following a recession, such as post the large recessions in the 70s, and after the great financial crisis. As can be seen in Figure 11, we observe that the Phillips Multiplier became particularly flat during the ZLB period, but after that has seen a steepening trend following the departure from the ZLB in the latter part of the sample.

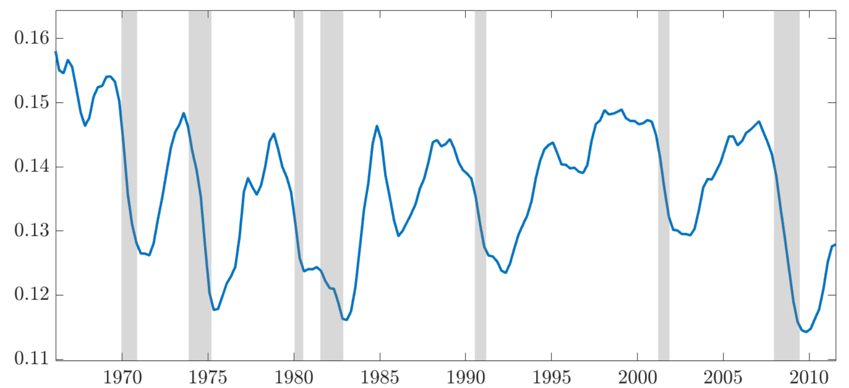
We compare the model-implied Phillips Multiplier in Figure 9a with the state-dependent local projection Phillips Multipliers of Figures 9b and 9c. Using unemployment as a state proxy as in

Figure 9: State-Dependence in the Phillips Multiplier

(a) Nonlinear model – nonlinear parameter estimates and states



(b) State-Dependent Local Projections with Romer and Romer (2004) shocks and Unemployment Rate proxy as state



(c) State-Dependent Local Projections with Romer and Romer (2004) shocks and Corporate Profit proxy as state

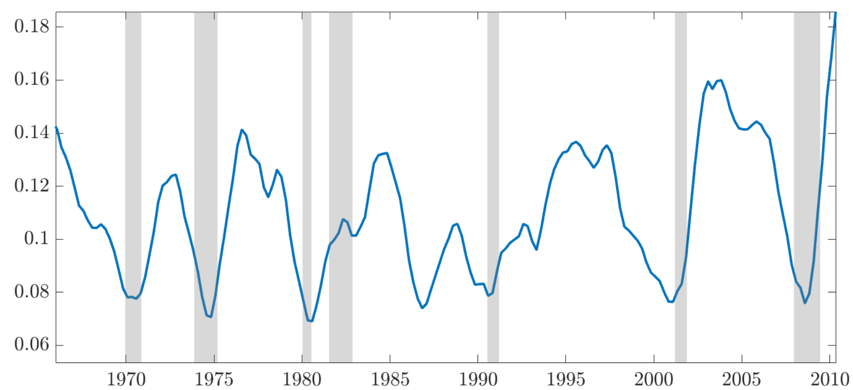
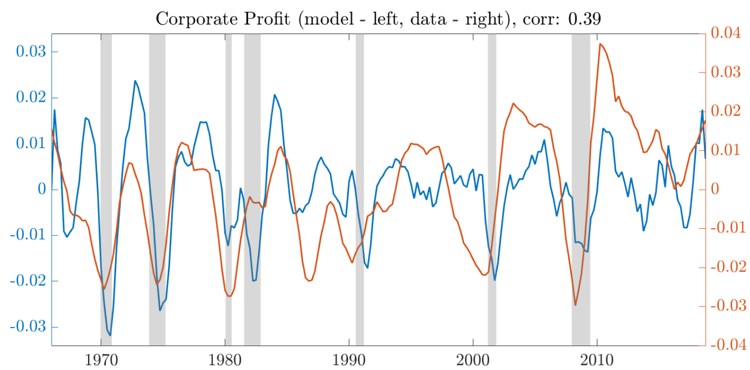


Figure 9b, we see that Figure 9b closely tracks the cyclicality in the Phillips Multiplier implied by the model, falling during recessions and rising during expansions.

Flow profit is not directly observed in the data, but based on the filtered the states, we can compare our corporate profit state-proxy implied by our model and compare this with the corporate profit data used in the empirical section. This serves as an external validity check. As shown in Figure 10, corporate profit in the model and data positively correlated with the data proxies, while being an untargeted moment. One thing to note is that the corporate profit data is leading the model, which explains why the state-dependent local projections in Figure 9c also suggest a Phillips Multiplier leading the model and the unemployment rate proxy-based local projection evidence.

Figure 10: Proxies for Surplus in Data and Through the Lens of the Model



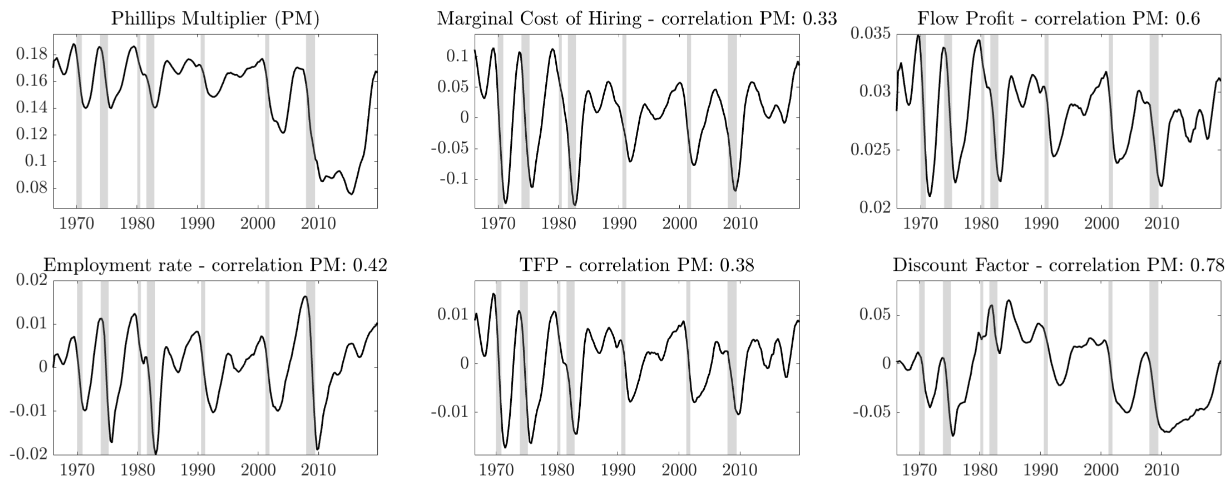
The fluctuations in the Phillips Multiplier can be explained and interpreted through the mechanisms outlined in this paper. To illustrate this, Figure 11 visualizes the Phillips Multiplier alongside relevant proxies and state variables, with their respective correlations noted. For example, the correlation between the model-implied marginal cost of hiring and the Phillips Multiplier across filtered states is 0.33, while its correlation with TFP is 0.38. Flow profit exhibits a stronger correlation of 0.6 with the Phillips Multiplier. Note that the Phillips Multiplier is also strongly influenced by the discount factor in this model in the latter part of the sample, in particular during the ZLB period.

Comparison with other empirical evidence on the Phillips Multiplier Looking at the existing literature, our average Phillips Multiplier estimates are consistent with Barnichon and Mesters (2021) post-1990's estimate.

7 Conclusion

This paper proposes a new methodology for the estimation of DSGE models featuring various nonlinearities, including state dependence. Our methodology contrasts the existing literature, which treats model solution and likelihood evaluation as two separate steps. Rather, we integrate the characterization of the model solution in the filtering step, and obtain an accurate and fast estimation procedure.

Figure 11: Phillips Multiplier alongside Surplus Proxies and Model States



The paper uses the Diamond-Mortensen-Pissarides model to explain and demonstrate the method, where the model is chosen for its known strong nonlinearities (Petrosky-Nadeau and Zhang, 2017). We then also illustrate the method in the textbook New Keynesian model, showing that we can estimate the model nonlinearly in ca. 1.5 times the amount of time it takes to estimate the model using linear methods, and many orders of magnitude faster than existing nonlinear approaches.

Next, in our main application, the paper establishes a link between time variation in the effects of monetary policy due to search frictions in the labor market. We first demonstrate this mechanism analytically, and show how flow surplus scales the response of variables to monetary policy shocks. Next, we use our estimation procedure to recover time-varying impulse response functions and Phillips Multipliers, revealing substantial variation in monetary policy effectiveness over the business cycle. Notably, we find that the Phillips Multiplier can fall to be about twice as small (in absolute values) during recessions as it is during expansions.

State dependence in the effects of monetary policy and the slope of the Phillips curve are critical considerations for policymakers. Our framework allows them to filter the underlying state of the economy and compute the implied impulse response functions and Phillips Multiplier in real time, ensuring that policy decisions are informed by current economic conditions rather than historical averages. More broadly, our modeling and filtering approach offers a generalizable framework for analyzing time-varying macroeconomic relationships. This methodology can be extended to study other forms of state dependence in policy transmission, or other relevant nonlinearities in DSGE models.

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Supplementary Appendix to “Efficient Estimation of Nonlinear DSGE Models”

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April, 2025

A Smoothing Algorithm

Algorithm 2 Time-Varying Kalman Smoother with Taylor Projections

Initialize: Set $t = T$, $r_T = 0$ and $N_T = 0$.

while $t \geq 1$ **do**

Update Step:

$$r_{t-1} = Z'_{\hat{s}_{t|t-1}} F_t^{-1} v_t + L'_t r_t \tag{A.1}$$

$$N_{t-1} = Z'_{\hat{s}_{t|t-1}} F_t^{-1} Z_{\hat{s}_{t|t-1}} + L'_t N_t L_t \tag{A.2}$$

where $L_t = T_{\hat{s}_{t|t-1}} - K_t Z_{\hat{s}_{t|t-1}}$, and K_t, v_t and F_t were stored from the Filter output.

Smoothing step:

Compute the smoothed state estimate:

$$\hat{s}_{t|T} = a_{t|t-1} + P_{t|t-1} r_{t-1} \tag{A.3}$$

Compute the smoothed covariance:

$$P_{t|T} = P_{t|t-1} - P_{t|t-1} N_{t-1} P_{t|t-1} \tag{A.4}$$

Set back time: Set $t \leftarrow t - 1$ and repeat.

end while

B Data

Table B1: Data details: Outcome Variables, Controls and State Proxies for Local Projections, Obtained January 9th, 2025.

Variable	Details	Transformation	Source
Unemployment Rate	Percent, Monthly, Seasonally Adjusted	None	LNS14000000
Real Wages	Average Hourly Earnings of Production and Nonsupervisory Employees, Total Private, Dollars per Hour, Monthly, Seasonally Adjusted	$100 \times \log(W/CPI)$	CES0500000008
Federal Funds Rate	Percent, Monthly, Not Seasonally Adjusted	None	DFF
Market tightness	Vacancy over unemployments	$100 \times \log$	Barnicon (2010) & FRED
Job finding rate		$100 \times \log$	Shimer (2005)
CPI	Consumer Price Index for All Urban Consumers: All Items in U.S. City Average, Index 1982-1984=100, Monthly, Seasonally Adjusted	$100 \times \log$	CPIAUCSL
PPI	Producer Price Index by Commodity: All Commodities, Index 1982=100, Monthly, Not Seasonally Adjusted	$100 \times \log$	PPIACO
IPI	Industrial Production: Total Index, Index 2017=100, Monthly, Seasonally Adjusted	$100 \times \log$	IP.B50001.S
HWI	Help-Wanted-Index from Barnicon (2010), JOLTS post 2001	$100 \times \log$	Barnicon (2010)
TFP	1 quarter moving average of TFP growth	Made into Index, Log, Hamilton Filter	Fernald (2014)
Corporate Profits	(Profit/Profit+Compensation)	log, Hamilton Filter	NIPA Table 1.10

Table B2: Data details: estimation and filtering

Variable	Details	Transformation	Source
Unemployment Rate	Percent, Quarterly, Seasonally Adjusted	Detrended*	UNRATE (FRED)
Average Weekly Earnings of Production and Nonsupervisory Employees Total Private	Dollars per Week, Quarterly, Seasonally Adjusted	Detrended* & demeaned	CES0500000030 (FRED)
Nonfarm Business Sector: Output per Worker	Index 2017=100 Quarterly Seasonally Adjusted	Detrended* & demeaned	PRS85006163
Market tightness	Vacancy over unemployments	Detrended*	Barnicon (2010) & FRED
Federal Funds Effective Rate	Percent	Quarterly**	DFF
CPI	Consumer Price Index for All Urban Consumers: All Items in U.S. City Average Index 1982-1984=100, Monthly, Seasonally Adjusted	Quarter-to-quarter % ch., detrended*	CPIAUCSL

Notes: (*) We use the Hamilton filter to remove long-run trends (Hamilton, 2018). (**) DFF converted to a quarterly interest rate ($100 * ((1 + DFF/100)^{(1/4)} - 1)$).

C Analytical log-linearization

C.1 Smaller version of the model

We first consider the small NKDMP model of Section 5.2.

Equilibrium Conditions

$$1 = \beta E_t \left[\frac{R_t}{\Pi_{t+1}} \left(\frac{C_{t+1}}{C_t} \right)^{-\tau} \right] \tag{C.1}$$

$$\frac{\omega_t}{\gamma} - 1 = -\phi(\Pi_t - \Pi)\Pi_t + \phi \mathbb{E}_t \left[Q_{t,t+1}(\Pi_{t+1} - \Pi)\Pi_{t+1} \frac{Y_{t+1}}{Y_t} \right] \tag{C.2}$$

$$C_t = Y_t = Z_t N_t \tag{C.3}$$

$$N_t = (1-\delta)N_{t-1} + q_t V_t \tag{C.4}$$

$$\ln Z_{t+1} = \rho_z \ln Z_t + \epsilon_{z,t+1} \tag{C.5}$$

$$q_t = \chi \left(1 + \theta_t^{\iota}\right)^{\frac{-1}{\iota}} \tag{C.6}$$

$$\theta_t = \frac{V_t + X_t}{1 - (1-\delta)N_{t-1}} \tag{C.7}$$

$$\frac{\kappa}{q_t} = (1-\omega_t)Z_t - w_t + (1-\delta)E_t \left[Q_{t,t+1} \frac{\kappa}{q_{t+1}} \right] \tag{C.8}$$

$$w_t = \eta(1-\omega_t)Z_t + (1-\eta)\nu + \eta(1-\delta)\kappa E_t \left[Q_{t,t+1}\theta_{t+1}\right] \tag{C.9}$$

$$\frac{R_t}{\bar{R}} = \left[\left(\frac{\Pi_t}{\bar{\Pi}} \right)^{\psi_1} \left(\frac{Y_t}{\bar{Y}} \right)^{\psi_2} \right]^{1-\rho_R} \left(\frac{R_{t-1}}{\bar{R}} \right)^{\rho_R} e^{\epsilon_{R,t}} \tag{C.10}$$

Steady State We consider a steady state with $\Pi = 1$.

$$\pi = 1 \tag{C.11}$$

$$R = \beta^{-1} \tag{C.12}$$

$$\omega = \gamma \tag{C.13}$$

$$Z = 1 \tag{C.14}$$

$$C = Y = N \tag{C.15}$$

$$\delta N = qV \tag{C.16}$$

$$\theta = \frac{V}{1-(1-\delta)N} \tag{C.17}$$

$$q(\theta) = \chi \left(1 + \theta^{\iota}\right)^{\frac{-1}{\iota}} \tag{C.18}$$

$$w = \eta\left[1 - \gamma + (1-\delta)\beta\kappa\theta\right] + (1-\eta)\nu \tag{C.19}$$

$$w = 1 - \gamma - \frac{\kappa}{q(\theta)}\left[1 - (1-\delta)\beta\right] \tag{C.20}$$

Log-Linear Solution

$$E_t\left[\hat{y}_{t+1}\right] = \hat{y}_t + \frac{1}{\tau}E_t\left[\hat{r}_t - \hat{\pi}_{t+1}\right] \tag{C.21}$$

$$\hat{c}_t = \hat{y}_t = \hat{z}_t + \hat{n}_t \tag{C.22}$$

$$\hat{n}_t = (1-\delta)\hat{n}_{t-1} + \delta\left(\hat{q}_t + \hat{v}_t\right) \tag{C.23}$$

$$\hat{z}_t = \rho_z \hat{z}_{t-1} + \epsilon_{z,t} \tag{C.24}$$

$$\hat{r}_t = (1-\rho_R)\left[\psi_1\hat{\pi}_t + \psi_2\hat{y}_t\right] + \rho_R\hat{r}_{t-1} + \epsilon_{R,t} \tag{C.25}$$

$$\hat{\pi}_t = \beta E_t\left[\hat{\pi}_{t+1}\right] - \frac{1}{\phi}\hat{\omega}_t \tag{C.26}$$

$$\hat{w}_t = \frac{(1-\gamma)\hat{z}_t - \gamma\hat{\omega}_t + (1-\delta)\kappa\beta\theta E_t\left[\hat{\pi}_{t+1} - \hat{r}_t + \hat{\theta}_{t+1}\right]}{1-\gamma+\frac{1-\eta}{\eta}\nu + (1-\delta)\kappa\beta\theta} \tag{C.27}$$

$$\hat{q}_t = \frac{-1}{1+\theta^{\iota}}\hat{\theta}_t \tag{C.28}$$

$$\hat{\theta}_t = \hat{v}_t + \frac{(1-\delta)N}{1-(1-\delta)N}\hat{n}_{t-1} \tag{C.29}$$

$$\begin{aligned}\hat{q}_t = {} & \frac{\gamma(1-\beta(1-\delta))}{1-\gamma-w}\hat{\omega}_t + \frac{(\gamma-1)(1-\beta(1-\delta))}{1-\gamma-w}\hat{z}_t + \frac{w(1-\beta(1-\delta))}{1-\gamma-w}\hat{w}_t \ldots \\ & + (1-\delta)\beta E_t[\hat{q}_{t+1} + \hat{r}_t - \hat{\pi}_{t+1}]\end{aligned} \tag{C.30}$$

These ten equations determine the dynamics of the ten variables $\left\{\hat{y}_t, \hat{r}_t, \hat{\pi}_t, \hat{z}_t, \hat{n}_t, \hat{q}_t, \hat{v}_t, \hat{w}_t, \hat{\omega}_t, \hat{\theta}_t\right\}$.

C.2 Quantitative Model Equilibrium

The first-order condition for bonds leads to the following Euler equation

$$1 = \beta\mathbb{E}_t\left[\left(\frac{\lambda_{t+1}}{\lambda_t}\right)^{-\tau}\frac{R_t}{\Pi_{t+1}}\right].$$

Optimal price setting and vacancy posting gives the following pricing equation

$$\frac{\omega_t}{\gamma} = 1 - \phi(\Pi_t - \bar{\Pi})\Pi_t + \phi\beta\mathbb{E}_t\left[\left(\frac{\lambda_{t+1}}{\lambda_t}\right)^{-\tau}\left(\frac{Y_{t+1}}{Y_t}\right)(\Pi_{t+1} - \bar{\Pi})\Pi_{t+1}\right].$$

The expression for the Nash wage and optimal vacancy posting decision by firms leads to the following job-creation condition

$$\frac{\kappa}{q_t} = (1-\eta)\left[(1-\omega_t)Z_t - \nu\right] + \beta(1-\delta)\mathbb{E}_t\left[\left(\frac{\lambda_{t+1}}{\lambda_t}\right)^{-\tau}\frac{\kappa}{q_{t+1}}\right]$$

To close the model, the interest rate rule and law of motion for employment are given by

$$
\begin{aligned}
R_t &= \zeta \log\left(e^{1/\zeta} + e^{\tilde{R}_t e^{m_t}/\zeta}\right) \\
\tilde{R}_t &= R\left(\frac{\Pi_t}{\bar{\Pi}}\right)^{\psi_\pi}\left(\frac{Y_t}{Y_{t-1}}\right)^{\psi_y} \\
N_t &= (1-\delta)N_{t-1} + \underbrace{q_t\theta_t\left[1-(1-\delta)N_{t-1}\right]}_{=\,q_t V_t}
\end{aligned}
$$

A minimum state variable representation of the model has five states $\mathbb{S}_t = \{N_{t-1}, Z_t, m_t, Z_{t-1}, D_t\}$ and three policies $\{\Pi(\mathbb{S}_t), \omega(\mathbb{S}_t), \theta(\mathbb{S}_t)\}$.

D Textbook New-Keynesian Model

D.1 Households

There is a representative infinitely lived household that maximizes

$$\mathbb{E}_0\left[\sum_{t=0}^{\infty}\beta^t D_t U\left(C_t, N_t\right)\right]$$

subject to a sequence of budget constraints

$$P_t C_t + B_t = R_{t-1}B_{t-1} + P_t W_t N_t + T_t$$

where C_t is consumption of the final good, B_t is an uncontingent nominal bond, R_{t-1} is the nominal interest rate, W_t is the real wage, and T_t contains dividends from firms net of taxes and transfers from the government. We assume utility has the following form

$$U\left(C_t, N_t\right) = \begin{cases} \frac{C_t^{1-\tau}-1}{1-\tau} - \chi\frac{N_t^{1+\eta}}{1+\eta}, \text{ if } \tau \neq 1 \\ \log(C_t) - \chi\frac{N_t^{1+\eta}}{1+\eta}, \text{ if } \tau = 1 \end{cases}$$

where $d_t = \log(D_t)$ is a random shock to the utility of consumption and follows an AR(1)

$$d_{t+1} = \rho_d d_t + \varepsilon_t^d, \quad \varepsilon_t^d \sim N(0, \sigma_d^2)$$

The optimal consumptions/saving and labor supply decisions are given by

$$\frac{\chi N_t^{\eta}}{C_t^{-\tau}} = W_t$$

$$1 = \beta\mathbb{E}_t\left[\frac{D_{t+1}}{D_t}\left(\frac{C_{t+1}}{C_t}\right)^{-\tau}\frac{R_t}{\Pi_{t+1}}\right]$$

where $\Pi_{t+1} = \frac{P_{t+1}}{P_t}$. The stochastic discount factor is defined as

$$Q_{t,t+1} = \beta\frac{D_{t+1}}{D_t}\left(\frac{C_{t+1}}{C_t}\right)^{-\tau}$$

D.2 Firms

Perfectly competitive final good firm combine a continuum of intermediate goods $j \in [0, 1]$.

Intermediate good j produced by a monopolist with technology $Y_t(j) = Z_t N_t(j)$ where Z_t is an aggregate technology process common to all firms. These firms choose prices $P_t(j)$ and labor inputs $N_t(j)$ to maximize the present discounted value of profit

$$\mathbb{E}_0 \left[Q_{0,t} \left\{ \frac{P_t(j)}{P_t} Z_t N_t(j) - W_t N_t(j) - \frac{\phi}{2} \left(\frac{P_t(j)}{P_{t-1}(j)} - \Pi \right)^2 Y_t \right\} \right]$$

subject to a sequence of constraints

$$Z_t N_t(j) = \left(\frac{P_t(j)}{P_t} \right)^{-1/\gamma} Y_t$$

Optimal choices of $N_t(j)$ and $P_t(j)$ lead to the following condition

$$\begin{aligned} 0 = &\frac{Z_t N_t(j)}{P_t} - \phi \left(\frac{P_t(j)}{P_{t-1}(j)} - \Pi \right) \frac{Y_t}{P_{t-1,j}} - \frac{1}{\gamma} \left(\frac{P_t(j)}{P_t} - \frac{W_t}{Z_t} \right) \left(\frac{P_t(j)}{P_t} \right)^{-1/\gamma - 1} \frac{Y_t}{P_t} \\ &+ \phi \mathbb{E}_t \left[Q_{t,t+1} \left(\frac{P_{t+1}(j)}{P_t(j)} - \Pi \right) \frac{P_{t+1}(j)}{P_t(j)^2} Y_{t+1} \right] \end{aligned}$$

In a symmetric equilibrium $P_t(j) = P_t$ and $Y_t = Z_t N_t(j)$ for all j so this simplifies to

$$\frac{1}{\gamma} \left(1 - \frac{W_t}{Z_t} \right) = 1 - \phi \left(\Pi_t - \Pi \right) \Pi_t + \phi \mathbb{E}_t \left[Q_{t,t+1} \left(\Pi_{t+1} - \Pi \right) \Pi_{t+1} \frac{Y_{t+1}}{Y_t} \right]$$

D.3 Government

The government levies lump-sum taxes and runs a balanced budget. The monetary authority follows a Taylor type rule

$$\frac{R_t}{R} = \left(\frac{\Pi_t}{\Pi} \right)^{\phi_\pi} \left(\frac{Y_t}{Y} \right)^{\phi_y} e^{m_t}$$

where m_t is an exogenous monetary policy shock that follows an AR(1)

$$m_{t+1} = \rho_m m_t + \varepsilon_t^m, \quad \varepsilon_t^m \sim N(0, \sigma_m^2)$$

D.4 Steady State

We focus on a non-stochastic steady state with inflation π equal to the Central Bank target which implies $R = \Pi/\beta$, and the parameter χ normalizes steady state output Y (that is N) to be unity $\chi = 1 - \gamma$. Note this implies $W = \chi$ so γ is the steady state wage markdown.

D.5 Equilibrium

The equilibrium conditions for market clearing in the labor and goods markets are

$$\begin{aligned} Y_t &= C_t = N_t Z_t \\ W_t &= (1-\gamma) N_t^{\eta} C_t^{\tau} = (1-\gamma) N_t^{\eta+\tau} Z_t^{\tau} \end{aligned}$$

A minimal state representation consists of states $\mathbb{S} = \{z_t, d_t, m_t\}$ and policies $\{\Pi(\mathbb{S}), N(\mathbb{S})\}$ defined through the equilibrium conditions (substituting market clearing conditions where appropriate)

$$\begin{aligned} \text{Euler/IS}: \ 1 &= \beta \mathbb{E}_t \left[e^{(\rho_d - 1) d_t + \epsilon^d_{t+1}} \left(\frac{N_{t+1}}{N_t} e^{(\rho_z - 1) z_t + \epsilon^z_{t+1}} \right)^{-\tau} \frac{R_t}{\Pi_{t+1}} \right] \\ \text{Pricing/NK-Phillips}: \ \frac{1}{\gamma} \left(1 - \chi N_t^{\eta+\tau} e^{(\tau-1) z_t} \right) &= 1 - \phi \left(\Pi_t - \Pi \right) \Pi_t \\ &+ \phi \beta \mathbb{E}_t \left[e^{(\rho_d - 1) d_t + \epsilon^d_{t+1}} \left(\frac{N_{t+1}}{N_t} e^{(\rho_z - 1) z_t + \epsilon^z_{t+1}} \right)^{1-\tau} \left(\Pi_{t+1} - \Pi \right) \Pi_{t+1} \right] \end{aligned}$$

and an auxiliary equation

$$\text{Taylor Rule}: \ \frac{R_t}{R} = \left(\frac{\Pi_t}{\Pi} \right)^{\phi_\pi} \left(\frac{N_t Z_t}{Y} \right)^{\phi_y} e^{m_t}$$

Using the steady state relations and letting lower case letters denote log-deviations from steady state values $x = \ln X_t - \ln X$, the equilibrium conditions become

$$\begin{aligned} 1 &= \mathbb{E}_t \left[e^{(\rho_d - 1) d_t + \varepsilon^d_{t+1} - \tau \left(n_{t+1} - n_t + (\rho_z - 1) z_t + \varepsilon^z_{t+1} \right) + r_t - \pi_{t+1}} \right] \\ \kappa \left(1 - e^{(\eta+\tau) n_t + (\tau - 1) z} \right) &= - \left(e^{\pi_t} - 1 \right) e^{\pi_t} + \beta \mathbb{E}_t \left[e^{(\rho_d - 1) d_t + \epsilon^d_{t+1} + (1-\tau) \left(n_{t+1} - n_t + (\rho_z - 1) z_t + \varepsilon^z_{t+1} \right)} \left(e^{\pi_{t+1}} - 1 \right) e^{\pi_{t+1}} \right] \\ r_t &= \phi_\pi \pi_t + \phi_y (n_t + z + t) + m_t \end{aligned}$$

where $\kappa = \frac{1-\gamma}{\phi \Pi^2 \gamma}$

D.6 Log-linear System

Lowercase signifies log-deviation from steady state

$$
\begin{aligned}
n_t &= \mathbb{E}_t[n_{t+1}] + \frac{1}{\tau}(1-\rho_d)d_t - (1-\rho_z)z_t - \frac{1}{\tau}\{r_t - \mathbb{E}_t[\pi_{t+1}]\} \\
\pi_t &= \beta\mathbb{E}_t[\pi_{t+1}] + \kappa\big[(\eta+\tau)n_t + (\tau-1)z_t\big] \\
r_t &= \phi_\pi \pi_t + \phi_y(n_t + z_t) + m_t \\
m_t &= \rho_m m_{t-1} + \varepsilon_t^m \\
z_t &= \rho_z z_{t-1} + \varepsilon_t^z \\
d_t &= \rho_d d_{t-1} + \varepsilon_t^d
\end{aligned}
$$

D.7 Local Approximation

Note, we can write the variables of the equilibrium system in terms of log deviations from steady state (lowercase signifies logs, e.g, $\pi_t = \ln \Pi_t - \ln \Pi$), after substituting the Taylor Rule into the Euler equation, as

$$
\begin{aligned}
1 =& \mathbb{E}_t\left[e^{(\rho_d-1)d_t+\epsilon_{t+1}^d-\tau\{n_{t+1}-n_t+(\rho_z-1)z_t+\epsilon_{t+1}^z\}+\phi_\pi\pi_t+\phi_y(n_t+z_t)+m_t-\pi_{t+1}}\right] \\
\frac{1-\gamma}{\Pi^2\phi\gamma}\left(1-e^{(\eta+\tau)n_t+(\tau-1)z_t}\right) =& -\left(e^{\pi_t}-1\right)e^{\pi_t} \\
&+\beta\mathbb{E}_t\left[e^{(\rho_d-1)d_t+\epsilon_{t+1}^d+(1-\tau)\{n_{t+1}-n_t+(\rho_z-1)z_t+\epsilon_{t+1}^z\}}\left(e^{\pi_{t+1}}-1\right)e^{\pi_{t+1}}\right]
\end{aligned}
$$

Which can be simplified to

$$
\begin{aligned}
1 =& e^{(\rho_d-1)d_t+\tau n_t+\tau(1-\rho_z)z_t+\phi_\pi\pi_t+\phi_y(n_t+z_t)+m_t}\mathbb{E}_t\left[e^{\epsilon_{t+1}^d-\tau n_{t+1}-\tau\epsilon_{t+1}^z-\pi_{t+1}}\right] \\
\frac{1-\gamma}{\Pi^2\phi\gamma}\left(1-e^{(\eta+\tau)n_t+(\tau-1)z_t}\right) =& -e^{2\pi_t}+e^{\pi_t} \\
&+\beta\mathbb{E}_t\left[e^{(\rho_d-1)d_t+\epsilon_{t+1}^d+(1-\tau)\{n_{t+1}-n_t+(\rho_z-1)z_t+\epsilon_{t+1}^z\}+2\pi_{t+1}}\right] \\
&-\beta\mathbb{E}_t\left[e^{(\rho_d-1)d_t+\epsilon_{t+1}^d+(1-\tau)\{n_{t+1}-n_t+(\rho_z-1)z_t+\epsilon_{t+1}^z\}+\pi_{t+1}}\right]
\end{aligned}
$$

The first step is to suppose the policies are linear in logs

$$
\begin{aligned}
\pi_t &= a_0 + a_1 z_t + a_2 d_t + a_3 m_t \\
n_t &= b_0 + b_1 z_t + b_2 d_t + b_3 m_t
\end{aligned}
$$

This implies the future policies are of the form

$$\begin{aligned}\pi_{t+1} &= a_0 + a_1\rho_z z_t + a_1\epsilon^z_{t+1} + a_2\rho_d d_t + a_2\epsilon^d_{t+1} + a_3\rho_m m_t + a_3\epsilon^m_{t+1}\\ n_{t+1} &= b_0 + b_1\rho_z z_t + b_1\epsilon^z_{t+1} + b_2\rho_d d_t + b_2\epsilon^d_{t+1} + b_3\rho_m m_t + b_3\epsilon^m_{t+1}\end{aligned}$$

Moreover, it implies

$$n_{t+1} - n_t = b_1(\rho_z - 1)z_t + b_1\epsilon^z_{t+1} + b_2(\rho_d - 1)d_t + b_2\epsilon^d_{t+1} + b_3(\rho_m - 1)m_t + b_3\epsilon^m_{t+1}$$

The goal of the local approximation method is to solve for the policy coefficients $\boldsymbol{a}$ and $\boldsymbol{b}$ as implicit functions of the current state (z_t, d_t, m_t).

Euler equation. Consider the Euler equation with the policies substituted

$$\begin{aligned}1 =& e^{(\rho_d-1)d_t-\tau\left[b_1(\rho_z-1)z_t+b_2(\rho_d-1)d_t+b_3(\rho_m-1)m_t\right]-\tau(\rho_z-1)z_t+r_t-a_1-a_1\rho_z z_t-a_2\rho_d d_t-a_3\rho_m m_t}\\ &\cdot \mathbb{E}_t\left[e^{\epsilon^d_{t+1}-\tau(b_1\epsilon^z_{t+1}+b_2\epsilon^d_{t+1}+b_3\epsilon^m_{t+1})-\tau\epsilon^z_{t+1}-a_1\epsilon^z_{t+1}-a_2\epsilon^d_{t+1}-a_3\epsilon^m_{t+1}}\right]\end{aligned}$$

Collecting terms in the expectation yields

$$\mathbb{E}_t\left[e^{\epsilon^d_{t+1}-\tau(b_1\epsilon^z_{t+1}+b_2\epsilon^d_{t+1}+b_3\epsilon^m_{t+1})-\tau\epsilon^z_{t+1}-a_1\epsilon^z_{t+1}-a_2\epsilon^d_{t+1}-a_3\epsilon^m_{t+1}}\right] = \mathbb{E}_t\left[e^{(1-\tau b_2-a_2)\epsilon^d_{t+1}-(\tau(b_1+1)+a_1)\epsilon^z_{t+1}-\tau(b_3+a_3)\epsilon^m_{t+1}}\right]$$

For a mean zero normal random variable X, and constant c, the expectation satisfies

$$\mathbb{E}\left[e^{cX}\right] = e^{\frac{1}{2}c^2\sigma_x^2},$$

so the Euler equation simplifies to

$$\begin{aligned}1 =& e^{(\rho_d-1)d_t-\tau\left[b_1(\rho_z-1)z_t+b_2(\rho_d-1)d_t+b_3(\rho_m-1)m_t\right]-\tau(\rho_z-1)z_t+r_t-a_1-a_1\rho_z z_t-a_2\rho_d d_t-a_3\rho_m m_t}\\ &\cdot e^{\frac{1}{2}\left\{(1-\tau b_2-a_2)^2\sigma_d^2+(\tau(b_1+1)+a_1)^2\sigma_z^2+\tau^2(b_3+a_3)^2\sigma_m^2\right\}}\end{aligned}$$

Note, by taking logs and substituting the Taylor Rule, this is equivalent to

$$\begin{aligned}0 =& (\rho_d - 1)d_t + \tau\left[b_1(1-\rho_z)z_t + b_2(1-\rho_d)d_t + b_3(1-\rho_m)m_t\right] + \tau(1-\rho_z)z_t\\ &+ \phi_\pi(a_0 + a_1 z_t + a_2 d_t + a_3 m_t) + \phi_y(b_0 + b_1 z_t + b_2 d_t + b_3 m_t + z_t) + m_t\\ &- (a_0 + a_1\rho_z z_t + a_2\rho_d d_t + a_3\rho_m m_t)\\ &+ \frac{1}{2}\left\{(1-\tau b_2 - a_2)^2\sigma_d^2 + (\tau(b_1+1)+a_1)^2\sigma_z^2 + \tau^2(b_3+a_3)^2\sigma_m^2\right\}\end{aligned}$$

which implies the variance terms show up in the Euler equation, but not the partial derivatives of the Euler equation with respect to the states. The partial derivatives w.r.t. the policy coefficients are given by

$$
\begin{aligned}
&\partial b_0 : \phi_y \\
&\partial b_1 : \left[\tau(1-\rho_z) + \phi_y\right] z_t + \tau\left(\tau(b_1+1) + a_1\right)\sigma_z^2 \\
&\partial b_2 : \left[\tau(1-\rho_d) + \phi_y\right] d_t - \tau\left(1 - \tau b_2 - a_2\right)\sigma_d^2 \\
&\partial b_3 : \left[\tau(1-\rho_m) + \phi_y\right] m_t + \tau^2\left(b_3 + a_3\right)\sigma_m^2 \\
&\partial a_0 : \phi_p - 1 \\
&\partial a_1 : (\phi_p - \rho_z) z_t + \left(\tau(b_1+1) + a_1\right)\sigma_z^2 \\
&\partial a_2 : (\phi_p - \rho_d) d_t - \left(1 - \tau b_2 - a_2\right)\sigma_d^2 \\
&\partial a_3 : (\phi_p - \rho_m) m_t + \tau^2\left(b_3 + a_3\right)\sigma_m^2
\end{aligned}
$$

Moreover, the partial derivatives with respect to the states are given by

$$
\begin{aligned}
\partial z_t &: 0 = \tau b_1 + \tau(1-\rho_z) + \phi_\pi a_1 + \phi_y b_1 - \tau b_1 \rho_z - a_1 \rho_z \\
&\Rightarrow 0 = \tau(1-\rho_z) + \left[\tau(1-\rho_z) + \phi_y\right] b_1 + (\phi_\pi - \rho_z) a_1 \\
\partial z_t \partial b_1 &: \tau(1-\rho_z) + \phi_y \\
\partial z_t \partial a_1 &: \phi_\pi - \rho_z
\end{aligned}
$$

$$
\begin{aligned}
\partial d_t &: 0 = (\rho_d - 1) + \tau b_2 + \phi_\pi a_2 + \phi_y b_2 - \tau b_2 \rho_d - a_2 \rho_d \\
&\Rightarrow 0 = (\rho_d - 1) + \left[\tau(1-\rho_d) + \phi_y\right] b_2 + (\phi_\pi - \rho_d) a_2 \\
\partial d_t \partial b_2 &: \tau(1-\rho_d) + \phi_y \\
\partial d_t \partial a_2 &: \phi_\pi - \rho_d
\end{aligned}
$$

$$
\begin{aligned}
\partial m_t &: 0 = \tau b_3 + \phi_\pi a_3 + \phi_y b_3 + 1 - \tau b_3 \rho_m - a_3 \rho_m \\
&\Rightarrow 0 = 1 + \left[\tau(1-\rho_m) + \phi_y\right] b_3 + (\phi_\pi - \rho_m) a_3 \\
\partial m_t \partial b_3 &: \tau(1-\rho_m) + \phi_y \\
\partial m_t \partial a_3 &: \phi_\pi - \rho_m
\end{aligned}
$$

which gives the residual function derivatives and elements of the Jacobian related to the Euler equation. Note, only the elements of the Jacobian that depend on the policy coefficients $\boldsymbol{a}$ and $\boldsymbol{b}$ need to be updated in the Newton solution step. Of the 64 elements of the Jacobian, 32 are determined by the Euler equation block, of which *only* 6 depend on the policy coefficients.

Zero Lower Bound. We are interested in models where there may be a binding zero lower bound on the nominal interest rate. Formally,

$$
\begin{aligned}
R_t &= \max\left\{1, R_t^s\right\} \\
\Rightarrow \frac{R_t}{R} &= \max\left\{\frac{1}{R}, \frac{R_t^s}{R}\right\} \\
\frac{R_t^s}{R} &= \left(\frac{\Pi_t}{\Pi}\right)^{\phi_\pi}\left(\frac{N_t Z_t}{Y}\right)^{\phi_y} e^{m_t}
\end{aligned}
$$

where R_t^s is the "shadow-rate" determined by a Taylor type rule. Taking log deviation from steady-state values $r_t = \ln R_t - \ln R$ and $r_t^s = \ln R_t^s - \ln R$, and denoting $r = \ln R$, we have

$$
\begin{aligned}
r_t &= \max\left\{-r, r_t^s\right\} \\
r_t^s &= \phi_\pi \pi_t + \phi_y(n_t + z_t) + m_t
\end{aligned}
$$

To deal with the ZLB we proceed by using a smooth approximation to the `max` operator. Specifically, we use the following

$$
r_t = \max\left\{-r, r_t^s\right\} \approx \frac{1}{\lambda}\ln\left(e^{-\lambda r} + e^{\lambda r_t^s}\right),
$$

where λ is a parameter. Note, as $\lambda \longrightarrow \infty$, the approximation becomes exact. Moreover, we have

$$
e^{r_t} = \left(e^{-\lambda r} + e^{\lambda r_t^s}\right)^{1/\lambda}
$$

Using the expression from before for the Euler equation we have

$$
\begin{aligned}
1 = &\left(e^{-\lambda r} + e^{\lambda\left(\phi_\pi(a_0 + a_1 z_t + a_2 d_t + a_3 m_t) + \phi_y(b_0 + b_1 z_t + b_2 d_t + b_3 m_t + z_t) + m_t\right)}\right)^{1/\lambda} \\
&e^{(\rho_d - 1)d_t - \tau\left[b_1(\rho_z - 1)z_t + b_2(\rho_d - 1)d_t + b_3(\rho_m - 1)m_t\right] - \tau(\rho_z - 1)z_t - a_1 - a_1\rho_z z_t - a_2\rho_d d_t - a_3\rho_m m_t} \\
&\cdot e^{\frac{1}{2}\left\{(1 - \tau b_2 - a_2)^2\sigma_d^2 + (\tau(b_1 + 1) + a_1)^2\sigma_z^2 + \tau^2(b_3 + a_3)^2\sigma_m^2\right\}},
\end{aligned}
$$

so we can again eliminate the expectation. The pricing equation is not directly affected by interest rates, so that block is unchanged.

Pricing equation. Now consider the pricing equation with the policies substituted

$$\frac{1-\gamma}{\Pi^2\phi\gamma}\left(1-e^{(\eta+\tau)(b_0+b_1z_t+b_2d_t+b_3m_t)+(\tau-1)z_t}\right)=-e^{2(a_0+a_1z_t+a_2d_t+a_3m_t)}+e^{a_0+a_1z_t+a_2d_t+a_3m_t}$$
$$+\beta e^{(\rho_d-1)d_t+(1-\tau)\left\{b_1(\rho_z-1)z_t+b_2(\rho_d-1)d_t+b_3(\rho_m-1)m_t+(\rho_z-1)z_t\right\}+2(a_0+a_1\rho_z z_t+a_2\rho_d d_t+a_3\rho_m m_t)}$$
$$\cdot\,\mathbb{E}_t\left[e^{\epsilon^d_{t+1}+(1-\tau)\left\{b_1\epsilon^z_{t+1}+b_2\epsilon^d_{t+1}+b_3\epsilon^m_{t+1}+\epsilon^z_{t+1}\right\}+2(a_1\epsilon^z_{t+1}+a_2\epsilon^d_{t+1}+a_3\epsilon^m_{t+1})}\right]$$
$$-\beta e^{(\rho_d-1)d_t+(1-\tau)\left\{b_1(\rho_z-1)z_t+b_2(\rho_d-1)d_t+b_3(\rho_m-1)m_t+(\rho_z-1)z_t\right\}+a_0+a_1\rho_z z_t+a_2\rho_d d_t+a_3\rho_m m_t}$$
$$\cdot\,\mathbb{E}_t\left[e^{\epsilon^d_{t+1}+(1-\tau)\left\{b_1\epsilon^z_{t+1}+b_2\epsilon^d_{t+1}+b_3\epsilon^m_{t+1}+\epsilon^z_{t+1}\right\}+a_1\epsilon^z_{t+1}+a_2\epsilon^d_{t+1}+a_3\epsilon^m_{t+1}}\right]$$

focusing on the expectation terms, we have the following

$$\mathbb{E}_t\left[e^{\epsilon^d_{t+1}+(1-\tau)\left\{b_1\epsilon^z_{t+1}+b_2\epsilon^d_{t+1}+b_3\epsilon^m_{t+1}+\epsilon^z_{t+1}\right\}+2(a_1\epsilon^z_{t+1}+a_2\epsilon^d_{t+1}+a_3\epsilon^m_{t+1})}\right]$$
$$=\mathbb{E}_t\left[e^{(1+(1-\tau)b_2+2a_2)\epsilon^d_{t+1}+((1-\tau)(b_1+1)+2a_1)\epsilon^z_{t+1}+((1-\tau)b_3+2a_3)\epsilon^m_{t+1}}\right]$$
$$=e^{\frac{1}{2}\left\{(1+(1-\tau)b_2+2a_2)^2\sigma_d^2+((1-\tau)(b_1+1)+2a_1)^2\sigma_z^2+((1-\tau)b_3+2a_3)^2\sigma_m^2\right\}}$$

$$\mathbb{E}_t\left[e^{\epsilon^d_{t+1}+(1-\tau)\left\{b_1\epsilon^z_{t+1}+b_2\epsilon^d_{t+1}+b_3\epsilon^m_{t+1}+\epsilon^z_{t+1}\right\}+a_1\epsilon^z_{t+1}+a_2\epsilon^d_{t+1}+a_3\epsilon^m_{t+1}}\right]$$
$$=\mathbb{E}_t\left[e^{(1+(1-\tau)b_2+a_2)\epsilon^d_{t+1}+((1-\tau)(b_1+1)+a_1)\epsilon^z_{t+1}+((1-\tau)b_3+a_3)\epsilon^m_{t+1}}\right]$$
$$=e^{\frac{1}{2}\left\{(1+(1-\tau)b_2+a_2)^2\sigma_d^2+((1-\tau)(b_1+1)+a_1)^2\sigma_z^2+((1-\tau)b_3+a_3)^2\sigma_m^2\right\}}$$

which implies the pricing equation simplifies to

$$\kappa\left(1-e^{(\eta+\tau)(b_0+b_1z_t+b_2d_t+b_3m_t)+(\tau-1)z_t}\right)=-e^{2(a_0+a_1z_t+a_2d_t+a_3m_t)}+e^{a_0+a_1z_t+a_2d_t+a_3m_t}$$
$$+\beta e^{(\rho_d-1)d_t+(1-\tau)\left\{b_1(\rho_z-1)z_t+b_2(\rho_d-1)d_t+b_3(\rho_m-1)m_t+(\rho_z-1)z_t\right\}+2(a_0+a_1\rho_z z_t+a_2\rho_d d_t+a_3\rho_m m_t)}$$
$$\cdot\,e^{\frac{1}{2}\left\{(1+(1-\tau)b_2+2a_2)^2\sigma_d^2+((1-\tau)(b_1+1)+2a_1)^2\sigma_z^2+((1-\tau)b_3+2a_3)^2\sigma_m^2\right\}}$$
$$-\beta e^{(\rho_d-1)d_t+(1-\tau)\left\{b_1(\rho_z-1)z_t+b_2(\rho_d-1)d_t+b_3(\rho_m-1)m_t+(\rho_z-1)z_t\right\}+a_0+a_1\rho_z z_t+a_2\rho_d d_t+a_3\rho_m m_t}$$
$$\cdot\,e^{\frac{1}{2}\left\{(1+(1-\tau)b_2+a_2)^2\sigma_d^2+((1-\tau)(b_1+1)+a_1)^2\sigma_z^2+((1-\tau)b_3+a_3)^2\sigma_m^2\right\}}$$

where $\kappa=\frac{1-\gamma}{\Pi^2\phi\gamma}$. Here, the derivatives of the residual function and elements of the Jacobian are more elaborate, but given that this is an analytic expression we can rely on symbolic differentiation to obtain the results.

E NK-DMP Model

Here we describe the NK-DMP model. The equilibrium conditions are

$$
\begin{aligned}
1 &= \beta \mathbb{E}_t \left[\frac{D_{t+1}}{D_t} \left(\frac{N_{t+1} Z_{t+1}}{N_t Z_t} \right)^{-\tau} \frac{R_t}{\Pi_{t+1}} \right] \\
\frac{\Omega_t}{\gamma} &= 1 - \phi \left(\Pi_t - \Pi \right) \Pi_t + \phi \beta \mathbb{E}_t \left[\frac{D_{t+1}}{D_t} \left(\frac{N_{t+1} Z_{t+1}}{N_t Z_t} \right)^{1-\tau} \left(\Pi_{t+1} - \Pi \right) \Pi_{t+1} \right] \\
\frac{c}{Q_t} &= (1-\eta) \left((1-\Omega_t) e^{z_t} - \nu \right) + \beta \mathbb{E}_t \left[(1-\delta_{t+1}) \frac{D_{t+1}}{D_t} \left(\frac{N_{t+1} Z_{t+1}}{N_t Z_t} \right)^{-\tau} \frac{c}{Q_{t+1}} \right]
\end{aligned}
$$

with auxiliary equations

$$
\begin{aligned}
\frac{R_t}{R} &= \left(\frac{\Pi_t}{\Pi} \right)^{\phi_\pi} \left(\frac{N_t Z_t}{N} \right)^{\phi_y} e^{m_t} \\
N_t &= (1-\delta_t) N_{t-1} + Q_t \Theta_t \left(1 - (1-\delta_t) N_{t-1} \right) \\
Q_t &= A_t \Theta_t^{-\alpha} \\
\delta_t &= \delta e^{s_t} \\
A_t &= A e^{a_t} \\
z_t &= \rho_z z_{t-1} + \epsilon_t^z \\
d_t &= \rho_d d_{t-1} + \epsilon_t^d \\
m_t &= \rho_m m_{t-1} + \epsilon_t^m \\
s_t &= \rho_s s_{t-1} + \epsilon_t^s \\
a_t &= \rho_a a_{t-1} + \epsilon_t^a
\end{aligned}
$$

where $z_t = \ln Z_t$ and $d_t = \ln D_t$. A minimal state representation has six states $\mathbb{S} = \{N_{t-1}, z_t, d_t, m_t, s_t, a_t\}$ and three policies $\{\theta(\mathbb{S}), \Pi(\mathbb{S}), \omega(\mathbb{S})\}$

E.1 Steady State

We focus on a non-stochastic steady state where inflation is at the central bank target Π. This implies $\beta = \Pi / R$ and $\omega = \gamma$. In the labor market, we normalize $\theta = 1$, and $q = 0.7$, and $N = 0.945$. This implies $A = 0.7$ and

$$
\delta N = q\theta - q\theta(1-\delta)N \Rightarrow \delta = \frac{1-N}{N} \frac{q\theta}{1-q\theta} \approx 0.1358
$$

E.2 Local Approximation

In what follows it will be useful to write variables X_t in terms of log deviations from their steady state values $x_t = \ln X_t - \ln X$. Moreover we will add a linear policy for employment n_t.

This yields an equilibrium system

$$
\begin{aligned}
1 &= \mathbb{E}_t \left[e^{d_{t+1}-d_t-\tau(n_{t+1}+z_{t+1}-n_t-z_t)+r_t-\pi_{t+1}} \right] \\
\frac{1}{\phi\Pi^2}\left(e^{\omega_t}-1\right) &= -\left(e^{\pi_t}-1\right)e^{\pi_t} + \beta\mathbb{E}_t\left[e^{d_{t+1}-d_t+(1-\tau)(n_{t+1}+z_{t+1}-n_t-z_t)}\left(e^{\pi_{t+1}}-1\right)e^{\pi_{t+1}} \right] \\
e^{-q_t} &= \frac{Q(1-\eta)}{c}\left((1-\gamma e^{\omega_t})e^{z_t}-\nu\right) + \beta\mathbb{E}_t\left[(1-\delta e^{s_{t+1}})e^{d_{t+1}-d_t-\tau(n_{t+1}+z_{t+1}-n_t-z_t)-q_{t+1}} \right] \\
e^{n_t-n_{t-1}} &= (1-\delta e^{s_t}) + Q\theta e^{q_t+\theta_t}\left(\frac{1}{N}e^{-n_{t-1}} - (1-\delta e^{s_t}) \right)
\end{aligned}
$$

where

$$
\begin{aligned}
\theta_t &= b_{10} + b_{11}n_{t-1} + b_{12}z_t + b_{13}d_t + b_{14}m_t + b_{15}s_t + b_{16}a_t \\
\pi_t &= b_{20} + b_{21}n_{t-1} + b_{22}z_t + b_{23}d_t + b_{24}m_t + b_{25}s_t + b_{26}a_t \\
\omega_t &= b_{30} + b_{31}n_{t-1} + b_{32}z_t + b_{33}d_t + b_{34}m_t + b_{35}s_t + b_{36}a_t \\
n_t &= b_{40} + b_{41}n_{t-1} + b_{42}z_t + b_{43}d_t + b_{44}m_t + b_{45}s_t + b_{46}a_t \\
q_t &= a_t - \alpha\theta_t \\
r_t &= \phi_\pi\pi_t + \phi_y(n_t+z_t) + m_t
\end{aligned}
$$

Which implies

$$
n_t - n_{t-1} = b_{40} + (b_{41}-1)n_{t-1} + b_{42}z_t + b_{43}d_t + b_{44}m_t + b_{45}s_t + b_{46}a_t
$$

$$
n_{t+1} - n_t = b_{41}(n_t - n_{t-1}) + b_{42}(z_{t+1}-z_t) + b_{43}(d_{t+1}-d_t) + b_{44}(m_{t+1}-m_t) + b_{45}(s_{t+1}-s_t) + b_{46}(a_{t+1}-a_t)
$$

$$
\begin{aligned}
&= b_{41}(n_t - n_{t-1}) + b_{42}(\rho_z-1)z_t + b_{43}(\rho_d-1)d_t + b_{44}(\rho_m-1)m_t + b_{45}(\rho_s-1)s_t + b_{46}(\rho_a-1)a_t \\
&\quad + b_{42}\varepsilon^z_{t+1} + b_{43}\varepsilon^d_{t+1} + b_{44}\varepsilon^m_{t+1} + b_{45}\varepsilon^s_{t+1} + b_{45}\varepsilon^a_{t+1}
\end{aligned}
$$

$$
\begin{aligned}
&= b_{41}(b_{40} + (b_{41}-1)n_{t-1} + b_{42}z_t + b_{43}d_t + b_{44}m_t + b_{45}s_t + b_{46}a_t) \\
&\quad + b_{42}(\rho_z-1)z_t + b_{43}(\rho_d-1)d_t + b_{44}(\rho_m-1)m_t + b_{45}(\rho_s-1)s_t + b_{46}(\rho_a-1)a_t \\
&\quad + b_{42}\varepsilon^z_{t+1} + b_{43}\varepsilon^d_{t+1} + b_{44}\varepsilon^m_{t+1} + b_{45}\varepsilon^s_{t+1} + b_{46}\varepsilon^a_{t+1}
\end{aligned}
$$

$$
\begin{aligned}
&= b_{41}b_{40} + b_{41}(b_{41}-1)n_{t-1} \\
&\quad + b_{42}(\rho_z-1+b_{41})z_t + b_{43}(\rho_d-1+b_{41})d_t + b_{44}(\rho_m-1+b_{41})m_t + b_{45}(\rho_s-1+b_{41})s_t + b_{46}(\rho_a-1+b_{41})a_t \\
&\quad + b_{42}\varepsilon^z_{t+1} + b_{43}\varepsilon^d_{t+1} + b_{44}\varepsilon^m_{t+1} + b_{45}\varepsilon^s_{t+1} + b_{46}\varepsilon^a_{t+1}
\end{aligned}
$$

Note, $n_{t+1} - n_t$ can be separated into two parts (i) terms determined at time t, and (ii) shocks that arrive at $t+1$.

$$n_{t+1} - n_t = B_0^n + B_1^n x_t + B_2^n \epsilon_{t+1}$$

Similarly, π_{t+1}, and the exogenous shocks can be written as

$$\begin{aligned}
\pi_{t+1} =& b_{20} + b_{21} n_t + b_{22} z_{t+1} + b_{23} d_{t+1} + b_{24} m_{t+1} + b_{25} s_{t+1} + b_{26} a_{t+1} \\
\\
=& b_{20} + b_{21}(b_{40} + b_{41} n_{t-1} + b_{42} z_t + b_{43} d_t + b_{44} m_t + b_{45} s_t + b_{46} a_t) \\
& + b_{22}\rho_z z_t + b_{23}\rho_d d_t + b_{24}\rho_m m_t + b_{25}\rho_s s_t + b_{26}\rho_a a_t \\
& + b_{22}\varepsilon^z_{t+1} + b_{23}\varepsilon^d_{t+1} + b_{24}\varepsilon^m_{t+1} + b_{25}\varepsilon^s_{t+1} + b_{26}\varepsilon^a_{t+1} \\
\\
=& B_0^\pi + B_1^\pi x_t + B_2^\pi \varepsilon_{t+1}
\end{aligned}$$

$$z_{t+1} - z_t = (\rho_z - 1) z_t + \varepsilon^z_{t+1}$$

$$d_{t+1} - d_t = (\rho_d - 1) d_t + \varepsilon^d_{t+1}$$

Finally θ_{t+1} and q_{t+1} can be written as

$$\begin{aligned}
\theta_{t+1} =& b_{10} + b_{11} n_t + b_{12} z_{t+1} + b_{13} d_{t+1} + b_{14} m_{t+1} + b_{15} s_{t+1} + b_{16} a_{t+1} \\
\\
=& b_{10} + b_{11}(b_{40} + b_{41} n_{t-1} + b_{42} z_t + b_{43} d_t + b_{44} m_t + b_{45} s_t + b_{46} a_t) \\
& + b_{12}\rho_z z_t + b_{13}\rho_d d_t + b_{14}\rho_m m_t + b_{15}\rho_s s_t + b_{16}\rho_a a_t \\
& + b_{12}\varepsilon^z_{t+1} + b_{13}\varepsilon^d_{t+1} + b_{14}\varepsilon^m_{t+1} + b_{15}\varepsilon^s_{t+1} + b_{16}\varepsilon^a_{t+1} \\
\\
=& B_0^\theta + B_1^\theta x_t + B_2^\theta \varepsilon_{t+1}
\end{aligned}$$

$$q_{t+1} = \rho_a a_t + \varepsilon^a_{t+1} - \alpha B_0^\theta - \alpha B_1^\theta x_t - \alpha B_2^\theta \varepsilon_{t+1}$$

Euler Equation. Consider the Euler equation

$$1 = \mathbb{E}_t \left[e^{d_{t+1} - d_t - \tau(n_{t+1} + z_{t+1} - n_t - z_t) + r_t - \pi_{t+1}} \right]$$

Using the previous expressions, we can write the Euler equation compactly as

$$1 = e^{(\rho_d - 1)d_t - \tau(\rho_z - 1)z_t - \tau B_0^n - \tau B_1^n x_t + r_t - B_0^\pi - B_1^\pi x_t} \mathbb{E}_t \left[e^{\varepsilon_{t+1}^d - \tau B_2^n \varepsilon_{t+1} - \tau \varepsilon_{t+1}^z - B_2^\pi \varepsilon_{t+1}} \right]$$

and the expectation can be expressed in closed form as

$$\begin{aligned}
&\mathbb{E}_t \left[e^{\varepsilon_{t+1}^d - \tau B_2^n \varepsilon_{t+1} - \tau \varepsilon_{t+1}^z - B_2^\pi \varepsilon_{t+1}} \right] \\
&= \mathbb{E}_t \left[e^{\varepsilon_{t+1}^d - \tau(b_{42}\varepsilon_{t+1}^z + b_{43}\varepsilon_{t+1}^d + b_{44}\varepsilon_{t+1}^m + b_{45}\varepsilon_{t+1}^s + b_{46}\varepsilon_{t+1}^a) - \tau\varepsilon_{t+1}^z - (b_{22}\varepsilon_{t+1}^z + b_{23}\varepsilon_{t+1}^d + b_{24}\varepsilon_{t+1}^m + b_{25}\varepsilon_{t+1}^s + b_{26}\varepsilon_{t+1}^a)} \right] \\
&= \mathbb{E}_t \left[e^{(1 - \tau b_{43} - b_{23})\varepsilon_{t+1}^d - (\tau(b_{42}+1) + b_{22})\varepsilon_{t+1}^z - (\tau b_{44} + b_{24})\varepsilon_{t+1}^m - (\tau b_{45} + b_{25})\varepsilon_{t+1}^s - (\tau b_{46} + b_{26})\varepsilon_{t+1}^a} \right] \\
&= e^{\frac{1}{2}\left((1 - \tau b_{43} - b_{23})^2 \sigma_d^2 + (\tau(b_{42}+1) + b_{22})^2 \sigma_z^2 + (\tau b_{44} + b_{24})^2 \sigma_m^2 + (\tau b_{45} + b_{25})^2 \sigma_s^2 + (\tau b_{46} + b_{26})^2 \sigma_a^2\right)}
\end{aligned}$$

Pricing equation Consider the optimal price setting condition

$$\frac{1}{\phi \Pi^2} \left(e^{\omega_t} - 1\right) = -\left(e^{\pi_t} - 1\right) e^{\pi_t} + \beta \mathbb{E}_t \left[e^{d_{t+1} - d_t + (1-\tau)(n_{t+1} + z_{t+1} - n_t - z_t)} \left(e^{\pi_{t+1}} - 1\right) e^{\pi_{t+1}} \right]$$

the continuation value can be solved in closed form given the linear policies, first note

$$\begin{aligned}
&\mathbb{E}_t \left[e^{d_{t+1} - d_t + (1-\tau)(n_{t+1} + z_{t+1} - n_t - z_t)} \left(e^{\pi_{t+1}} - 1\right) e^{\pi_{t+1}} \right] \\
&= \mathbb{E}_t \left[e^{d_{t+1} - d_t + (1-\tau)(n_{t+1} + z_{t+1} - n_t - z_t)} \left(e^{2\pi_{t+1}} - e^{\pi_{t+1}}\right) \right] \\
&= e^{(\rho_d - 1)d_t + (1-\tau)\left[(\rho_z - 1)z_t + B_0^n + B_1^n x_t\right]} \mathbb{E}_t \left[e^{\varepsilon_{t+1}^d + (1-\tau)(B_2^n \varepsilon_{t+1} + \varepsilon_{t+1}^z)} \left(e^{2\pi_{t+1}} - e^{\pi_{t+1}}\right) \right]
\end{aligned}$$

and the remaining expectation term can be written as

$$\mathbb{E}_t \left[e^{\varepsilon_{t+1}^d + (1-\tau)(B_2^n \varepsilon_{t+1} + \varepsilon_{t+1}^z) + 2\pi_{t+1}} \right] - \mathbb{E}_t \left[e^{\varepsilon_{t+1}^d + (1-\tau)(B_2^n \varepsilon_{t+1} + \varepsilon_{t+1}^z) + \pi_{t+1}} \right]$$

where

$$\begin{aligned}
\mathbb{E}_t \left[e^{\varepsilon_{t+1}^d + (1-\tau)(B_2^n \varepsilon_{t+1} + \varepsilon_{t+1}^z) + 2\pi_{t+1}} \right] &= e^{2B_0^\pi + 2B_1^\pi x_t} \mathbb{E}_t \left[e^{\varepsilon_{t+1}^d + (1-\tau)(B_2^n \varepsilon_{t+1} + \varepsilon_{t+1}^z) + 2B_2^\pi \varepsilon_{t+1}} \right] \\
\mathbb{E}_t \left[e^{\varepsilon_{t+1}^d + (1-\tau)(B_2^n \varepsilon_{t+1} + \varepsilon_{t+1}^z) + \pi_{t+1}} \right] &= e^{B_0^\pi + B_1^\pi x_t} \mathbb{E}_t \left[e^{\varepsilon_{t+1}^d + (1-\tau)(B_2^n \varepsilon_{t+1} + \varepsilon_{t+1}^z) + B_2^\pi \varepsilon_{t+1}} \right]
\end{aligned}$$

expanding the expectation terms and solving yields

$$
\begin{aligned}
&\mathbb{E}_t\left[e^{\varepsilon^d_{t+1}+(1-\tau)(B^n_2\varepsilon_{t+1}+\varepsilon^z_{t+1})+2B^\pi_2\varepsilon_{t+1}}\right]\\
&=\mathbb{E}_t\left[e^{\varepsilon^d_{t+1}+(1-\tau)(b_{42}\varepsilon^z_{t+1}+b_{43}\varepsilon^d_{t+1}+b_{44}\varepsilon^m_{t+1}+b_{45}\varepsilon^s_{t+1}+b_{46}\varepsilon^a_{t+1}+\varepsilon^z_{t+1})+2\left(b_{22}\varepsilon^z_{t+1}+b_{23}\varepsilon^d_{t+1}+b_{24}\varepsilon^m_{t+1}+b_{25}\varepsilon^s_{t+1}+b_{26}\varepsilon^a_{t+1}\right)}\right]\\
&=\mathbb{E}_t\left[e^{(1+(1-\tau)b_{43}+2b_{23})\varepsilon^d_{t+1}+((1-\tau)(b_{42}+1)+2b_{22})\varepsilon^z_{t+1}+((1-\tau)b_{44}+2b_{24})\varepsilon^m_{t+1}+((1-\tau)b_{45}+2b_{25})\varepsilon^s_{t+1}+((1-\tau)b_{46}+2b_{26})\varepsilon^a_{t+1}}\right]\\
&=e^{\frac{1}{2}\left((1+(1-\tau)b_{43}+2b_{23})^2\sigma^2_d+((1-\tau)(b_{42}+1)+2b_{22})^2\sigma^2_z+((1-\tau)b_{44}+2b_{24})^2\sigma^2_m+((1-\tau)b_{45}+2b_{25})^2\sigma^2_s+((1-\tau)b_{46}+2b_{26})^2\sigma^2_a\right)}
\end{aligned}
$$

$$
\begin{aligned}
&\mathbb{E}_t\left[e^{\varepsilon^d_{t+1}+(1-\tau)(B^n_2\varepsilon_{t+1}+\varepsilon^z_{t+1})+B^\pi_2\varepsilon_{t+1}}\right]\\
&=\mathbb{E}_t\left[e^{\varepsilon^d_{t+1}+(1-\tau)(b_{42}\varepsilon^z_{t+1}+b_{43}\varepsilon^d_{t+1}+b_{44}\varepsilon^m_{t+1}+b_{45}\varepsilon^s_{t+1}+b_{46}\varepsilon^a_{t+1}+\varepsilon^z_{t+1})+b_{22}\varepsilon^z_{t+1}+b_{23}\varepsilon^d_{t+1}+b_{24}\varepsilon^m_{t+1}+b_{25}\varepsilon^s_{t+1}+b_{26}\varepsilon^a_{t+1}}\right]\\
&=\mathbb{E}_t\left[e^{(1+(1-\tau)b_{43}+b_{23})\varepsilon^d_{t+1}+((1-\tau)(b_{42}+1)+b_{22})\varepsilon^z_{t+1}+((1-\tau)b_{44}+b_{24})\varepsilon^m_{t+1}+((1-\tau)b_{45}+b_{25})\varepsilon^s_{t+1}+((1-\tau)b_{46}+b_{26})\varepsilon^a_{t+1}}\right]\\
&=e^{\frac{1}{2}\left((1+(1-\tau)b_{43}+b_{23})^2\sigma^2_d+((1-\tau)(b_{42}+1)+b_{22})^2\sigma^2_z+((1-\tau)b_{44}+b_{24})^2\sigma^2_m+((1-\tau)b_{45}+b_{25})^2\sigma^2_s+((1-\tau)b_{46}+b_{26})^2\sigma^2_a\right)}
\end{aligned}
$$

Job creation condition. Consider the job creation condition

$$e^{-q_t}=\frac{Q(1-\eta)}{c}\left((1-\gamma e^{\omega_t})e^{z_t}-\nu\right)+\beta\mathbb{E}_t\left[(1-\delta e^{s_{t+1}})e^{d_{t+1}-d_t-\tau(n_{t+1}+z_{t+1}-n_t-z_t)-q_{t+1}}\right]$$

the continuation value can be solved in closed form given the linear policies, first note

$$
\begin{aligned}
&\mathbb{E}_t\left[(1-\delta e^{s_{t+1}})e^{d_{t+1}-d_t-\tau(n_{t+1}+z_{t+1}-n_t-z_t)-q_{t+1}}\right]\\
&=\mathbb{E}_t\left[(1-\delta e^{s_{t+1}})e^{d_{t+1}-d_t-\tau\left(B^n_0+B^n_1x_t+B^n_2\epsilon_{t+1}+z_{t+1}-z_t\right)-\rho_a a_t-\varepsilon^a_{t+1}+\alpha B^\theta_0+\alpha B^\theta_1 x_t+\alpha B^\theta_2\varepsilon_{t+1}}\right]\\
&=e^{(\rho_d-1)d_t-\tau(\rho_z-1)z_t-\tau B^n_0-\tau B^n_1 x_t-\rho_a a_t+\alpha B^\theta_0+\alpha B^\theta_1 x_t}\mathbb{E}_t\left[(1-\delta e^{s_{t+1}})e^{\varepsilon^d_{t+1}-\tau\left(B^n_2\epsilon_{t+1}+\varepsilon^z_{t+1}\right)-\varepsilon^a_{t+1}+\alpha B^\theta_2\varepsilon_{t+1}}\right]
\end{aligned}
$$

The remaining expectation term can be written as

$$
\begin{aligned}
&\mathbb{E}_t\left[(1-\delta e^{s_{t+1}})e^{\varepsilon^d_{t+1}-\tau\left(B^n_2\epsilon_{t+1}+\varepsilon^z_{t+1}\right)-\varepsilon^a_{t+1}+\alpha B^\theta_2\varepsilon_{t+1}}\right]\\
&=\mathbb{E}_t\left[e^{\varepsilon^d_{t+1}-\tau\left(B^n_2\epsilon_{t+1}+\varepsilon^z_{t+1}\right)-\varepsilon^a_{t+1}+\alpha B^\theta_2\varepsilon_{t+1}}\right]\\
&-\delta e^{\rho_s s_t}\mathbb{E}_t\left[e^{\varepsilon^s_{t+1}+\varepsilon^d_{t+1}-\tau\left(B^n_2\epsilon_{t+1}+\varepsilon^z_{t+1}\right)-\varepsilon^a_{t+1}+\alpha B^\theta_2\varepsilon_{t+1}}\right]
\end{aligned}
$$

where

$$\mathbb{E}_t\left[e^{\varepsilon^d_{t+1}-\tau\left(B^n_2\epsilon_{t+1}+\varepsilon^z_{t+1}\right)-\varepsilon^a_{t+1}+\alpha B^\theta_2\varepsilon_{t+1}}\right]$$
$$=\mathbb{E}_t\left[e^{\varepsilon^d_{t+1}-\tau\left(b_{42}\varepsilon^z_{t+1}+b_{43}\varepsilon^d_{t+1}+b_{44}\varepsilon^m_{t+1}+b_{45}\varepsilon^s_{t+1}+b_{46}\varepsilon^a_{t+1}+\varepsilon^z_{t+1}\right)-\varepsilon^a_{t+1}+\alpha\left(b_{12}\varepsilon^z_{t+1}+b_{13}\varepsilon^d_{t+1}+b_{14}\varepsilon^m_{t+1}+b_{15}\varepsilon^s_{t+1}+b_{16}\varepsilon^a_{t+1}\right)}\right]$$
$$=\mathbb{E}_t\left[e^{(\alpha b_{13}-\tau b_{43}+1)\varepsilon^d_{t+1}+(\alpha b_{12}-\tau b_{42})\varepsilon^z_{t+1}+(\alpha b_{14}-\tau b_{44})\varepsilon^m_{t+1}+(\alpha b_{15}-\tau b_{45})\varepsilon^s_{t+1}+(\alpha b_{16}-\tau b_{46}-1)\varepsilon^a_{t+1}}\right]$$
$$=e^{\frac{1}{2}\left((\alpha b_{13}-\tau b_{43}+1)^2\sigma^2_d+(\alpha b_{12}-\tau b_{42})^2\sigma^2_z+(\alpha b_{14}-\tau b_{44})^2\sigma^2_m+(\alpha b_{15}-\tau b_{45})^2\sigma^2_s+(\alpha b_{16}-\tau b_{46}-1)^2\sigma^2_a\right)}$$

$$\mathbb{E}_t\left[e^{\varepsilon^s_{t+1}+\varepsilon^d_{t+1}-\tau\left(B^n_2\epsilon_{t+1}+\varepsilon^z_{t+1}\right)-\varepsilon^a_{t+1}+\alpha B^\theta_2\varepsilon_{t+1}}\right]$$
$$=\mathbb{E}_t\left[e^{\varepsilon^s_{t+1}+\varepsilon^d_{t+1}-\tau\left(b_{42}\varepsilon^z_{t+1}+b_{43}\varepsilon^d_{t+1}+b_{44}\varepsilon^m_{t+1}+b_{45}\varepsilon^s_{t+1}+b_{46}\varepsilon^a_{t+1}+\varepsilon^z_{t+1}\right)-\varepsilon^a_{t+1}+\alpha\left(b_{12}\varepsilon^z_{t+1}+b_{13}\varepsilon^d_{t+1}+b_{14}\varepsilon^m_{t+1}+b_{15}\varepsilon^s_{t+1}+b_{16}\varepsilon^a_{t+1}\right)}\right]$$
$$=\mathbb{E}_t\left[e^{(\alpha b_{13}-\tau b_{43}+1)\varepsilon^d_{t+1}+(\alpha b_{12}-\tau b_{42})\varepsilon^z_{t+1}+(\alpha b_{14}-\tau b_{44})\varepsilon^m_{t+1}+(\alpha b_{15}-\tau b_{45}+1)\varepsilon^s_{t+1}+(\alpha b_{16}-\tau b_{46}-1)\varepsilon^a_{t+1}}\right]$$
$$=e^{\frac{1}{2}\left((\alpha b_{13}-\tau b_{43}+1)^2\sigma^2_d+(\alpha b_{12}-\tau b_{42})^2\sigma^2_z+(\alpha b_{14}-\tau b_{44})^2\sigma^2_m+(\alpha b_{15}-\tau b_{45}+1)^2\sigma^2_s+(\alpha b_{16}-\tau b_{46}-1)^2\sigma^2_a\right)}$$

so the expectation term becomes

$$\mathbb{E}_t\left[(1-\delta e^{s_{t+1}})e^{\varepsilon^d_{t+1}-\tau\left(B^n_2\epsilon_{t+1}+\varepsilon^z_{t+1}\right)-\varepsilon^a_{t+1}+\alpha B^\theta_2\varepsilon_{t+1}}\right]$$
$$=\left(e^{\frac{1}{2}(\alpha b_{15}-\tau b_{45})^2\sigma^2_s}-\delta e^{\rho_s s_t+\frac{1}{2}(\alpha b_{15}-\tau b_{45}+1)^2\sigma^2_s}\right)$$
$$\cdot\, e^{\frac{1}{2}\left((\alpha b_{13}-\tau b_{43}+1)^2\sigma^2_d+(\alpha b_{12}-\tau b_{42})^2\sigma^2_z+(\alpha b_{14}-\tau b_{44})^2\sigma^2_m+(\alpha b_{15}-\tau b_{45}+1)^2\sigma^2_s+(\alpha b_{16}-\tau b_{46}-1)^2\sigma^2_a\right)}$$

Law of motion for employment. Taking the law of motion for employment and substituting the expressions for the policies yields

$$e^{b_{40}+(b_{41}-1)n_{t-1}+b_{42}z_t+b_{43}d_t+b_{44}m_t+b_{45}s_t+b_{46}a_t}=$$
$$1-\delta e^{s_t}+Q\theta e^{a_t+(1-\alpha)(b_{10}+b_{11}n_{t-1}+b_{12}z_t+b_{13}d_t+b_{14}m_t+b_{15}s_t+b_{16}a_t)}\left(\frac{e^{-n_{t-1}}}{N}-1+\delta e^{s_t}\right)$$

F Comparing EKF of the true global solution and Taylor Projection

$$\left(g\circ h\right)=g-g'\cdot k+g'\cdot(h-h'\cdot k)+g'\cdot h'\cdot k$$

Thus, assuming linearity imposes two additional restrictions so the system is determinate. Now suppose we take a first-order approximation of the true policy functions around k_t

$$
\begin{aligned}
\hat{g}(k) &= g(k_t) + g'(k_t)(k - k_t) \\
\hat{h}(k) &= h(k_t) + h'(k_t)(k - k_t)
\end{aligned}
$$

Which implies

$$
\begin{aligned}
\hat{c}_t|_{k=k_t} &= \hat{g}(k_t) = g(k_t) = c_t \\
\hat{k}_{t+1}|_{k=k_t} &= \hat{h}(k_t) = h(k_t) = k_{t+1} \\
\hat{c}_{t+1}|_{k=k_t} &= \hat{g}(\hat{h}(k_t)) \\
&= g(k_t) + g'(k_t)(h(k_t) - k_t)
\end{aligned}
$$

If this were to correspond to the TP solution, we would have

$$
\begin{aligned}
a_0 &= g(k_t) - g'(k_t)k_t \\
a_1 &= g'(k_t) \\
b_0 &= h(k_t) - g'(k_t)k_t \\
b_1 &= h'(k_t) \\
a_0 + a_1 b_1 &= g(k_t) + g'(k_t)h(k_t) \\
a_1 b_1 &= g'(k_t)h(k_t)
\end{aligned}
$$

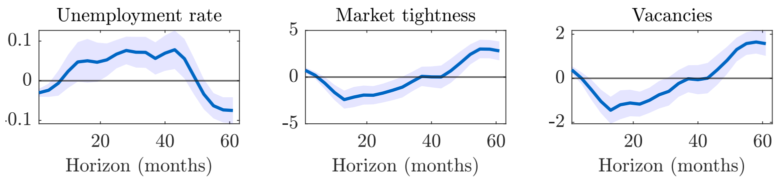
If $k_t \neq 0$, the condition only holds if $h(k_t) = h'(k_t)$ and $g'(k_t) = 0$. Note that Taylor Projection (TP) and a Taylor expansion of the true policy rules coincide at the non-stochastic steady state, where $k_t = 0$. By construction, Taylor Projection at k_t ensures that Euler errors are exactly zero at k_t. In contrast, for a Taylor expansion of the true policy, Euler errors at k_t are generally nonzero.

The main differences between the two approaches arise from the composition $g(h(k_t))$. Under TP, the composition determines the coefficients by solving the system of optimality conditions. In a Taylor expansion of the true policy, the coefficients are determined by the expansion itself and generally do not satisfy the optimality conditions. In particular, the composition of the Taylor expansion at k_t differs from the composition of the true solution evaluated at k_t.

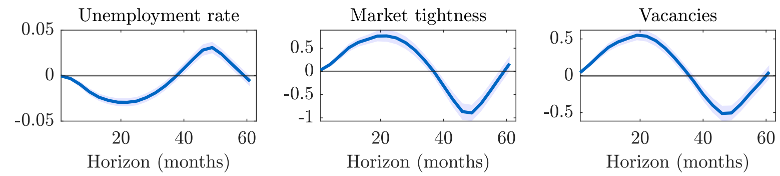
Although it may seem that the Implicit Function Theorem could be applied to equate the two approaches, this is not valid: the Taylor expansion does not satisfy the necessary optimality conditions, and hence the assumptions of the Implicit Function Theorem are not met.

G Supporting Empirical Evidence

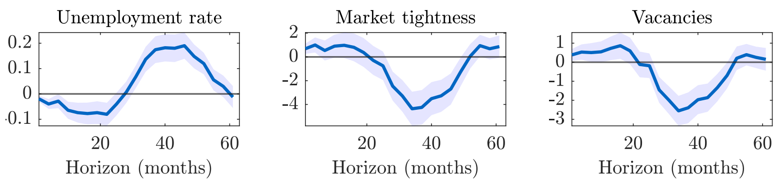
(a) Local Projection Baseline – Romer and Romer (2004) shock, Corporate Profit Proxy, 90% CI



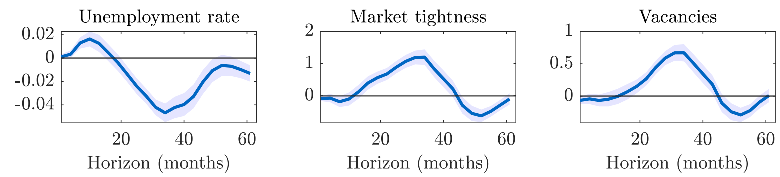
(b) Local Projection Interaction Term – Romer and Romer (2004) shock, Corporate Profit Proxy, 90% CI



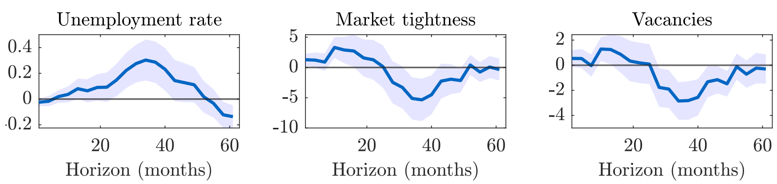
(c) Local Projection Baseline – Aruoba and Drechsel (2024) shock, Corporate Profit Proxy, 90% CI



(d) Local Projection Interaction Term – Aruoba and Drechsel (2024) shock, Corporate Profit Proxy, 90% CI



(e) Local Projection Baseline – Bauer and Swanson (2023) shock, Corporate Profit Proxy, 90% CI



(f) Local Projection Interaction Term – Bauer and Swanson (2023) shock, Corporate Profit Proxy, 90% CI